

Advanced Group Theory

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April 28, 2026

Abstract

Leading off from Groups and Rings, these notes deal with the more advanced objects of study in the theory of groups. In particular, we will focus on finite group theory. However, in the first half we present useful result on other topics of mathematics that rely on group theory such as algebraic topology and representation theory.

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1 Free Groups and Free Product of Groups

1.1 Free Groups

Definition 1.1.1 (The Words of a Set)

Let S be a set. Let $T = S \times \{-1, 1\}$. A word of S of length $n \in \mathbb{N} \setminus \{0\}$ is a map of sets $\omega : \{1, \dots, n\} \rightarrow T$. Such a word is denoted by

$$s_1 \cdots s_n$$

where $s_k = \omega(k)$. Define the empty word to be the unique map $\emptyset \rightarrow T$.

We think of $S \times \{1\}$ as the original copy of S , and $S \times \{-1\}$ as formally adjoining inverse elements so that we can form a group.

Definition 1.1.2 (Reduced Words)

Let S be a set. Define the projection maps $p_S : S \times \{-1, 1\} \rightarrow S$ and $p_{\{-1, 1\}} : S \times \{-1, 1\} \rightarrow \{-1, 1\}$ by $p_S(s, i) = s$ and $p_{\{-1, 1\}}(s, i) = i$ respectively. Let $s = s_1 \cdots s_n$ be a word of length n . We say that s is reduced if for $1 \leq i < n$, $p_S(s_i) = p_S(s_{i+1})$ implies that $p_{\{-1, 1\}}(s_i) \neq p_{\{-1, 1\}}(s_{i+1})$. By convention the empty word is reduced.

Intuitively, since $(s, -1)$ for $s \in S$ represents the inverse of $(s, 1)$, saying that $s_1 \cdots s_n$ is reduced means that no element is next to its inverse.

Definition 1.1.3 (Set of Reduced Words)

Let S be a set. Define the set of reduced words by

$$F_S = \{w \mid w \text{ is a reduced word of } S\}$$

Proposition 1.1.4

Let S be a set. Recursively define a product $*$: $F_S \times F_S \rightarrow F_S$ on the set of reduced words as follows

$$(s_1 \cdots s_m) * (t_1 \cdots t_n) = \begin{cases} s_1 \cdots s_{m-1} * t_2 \cdots t_n & \text{if } p_S(s_m) = p_S(t_1) \text{ and } p_{\{0, 1\}}(s_m) \neq p_{\{0, 1\}}(t_1) \\ s_1 \cdots s_m t_1 \cdots t_n & \text{otherwise} \end{cases}$$

Then F_S is a group whose binary operation is given by the above product.

Proof

We clearly have the identity 1 and the inverse of $g = g_1 \cdots g_n$ to be $g_n^{-1} \cdots g_1^{-1}$. We just have to check associativity and closure. By definition of the free product, the free product of reduced words is again a reduced word by definition and finiteness of the length of a reduced word.

For any $g \in G_i$, define the action of left multiplication by $L_g(h) = g * h$ for any $h \in *_i G_i$. We clearly have $L_{g_1} \circ L_{g_2} = L_{g_1 g_2}$ and $L_{g^{-1}} = L_g^{-1}$ and thus $L_g \in \text{Sym}(*_i G_i)$. For any word $g = g_1 \cdots g_m$, the map $L : *_i G_i \rightarrow \text{Sym}(*_i G_i)$ is injective since for any $g \in *_i G_i$, we have $L_g(1) = g$ which means that $g \neq h$ implies $L_g \neq L_h$. Thus we have embedded the set into the group $\text{Sym}(*_i G_i)$ so that associativity follows. ■

Definition 1.1.5 (The Free Group on a Set)

Let S be a set. Define the free group on S to be the set of reduced words

$$F_S = \{w \mid w \text{ is a reduced word of } S\}$$

together with the binary operation given by the above product.

Proposition 1.1.6 (Universal Property of Free Groups)

Let S be a set. Then the free group F_S and the inclusion map $\iota : S \rightarrow F_S$ satisfies the following universal property. For any group G and any map of sets $\psi : S \rightarrow G$, there exists a unique group homomorphism $\phi : F_S \rightarrow G$ such that the following diagram commutes:

$$\begin{array}{ccc} S & \xrightarrow{\iota} & F_S \\ & \searrow \psi & \downarrow \exists! \phi \\ & & G \end{array}$$

Moreover, F_S is the unique group (up to unique isomorphism) that has such a property.

Proposition 1.1.7 Let S, T be sets. Then $F_S \cong F_T$ as groups if and only if S and T are in bijection.

Lemma 1.1.8

Let S be a set. Then S is a generating set of the free group F_S on S .

Definition 1.1.9 (Free Group of Rank n)

Let $n \in \mathbb{N}$. Define the free group of rank n to be the free group F_n on the set $\{1, \dots, n\}$.

Proposition 1.1.10

Let G be a group. Then there exists a set S and a normal subgroup N of F_S such that there is a group isomorphism $G \cong F_S/N$.

1.2 Free Products of Groups

Definition 1.2.1 (The Words of a Collection of Groups)

Let $\{G_i \mid i \in I\}$ be a collection of groups. A word on these groups is a finite sequence $g_1 \cdots g_m$ where $g_k \in G_i$ for some $i \in I$. We say that such a word has length m .

Definition 1.2.2 (Reduced Words)

Let $\{G_i \mid i \in I\}$ be a collection of groups. Let $g_1 \cdots g_m$ be a word. The word is said to be reduced if the following are true.

- $g_k \neq 1_i$ for any $i \in I$ and for $k \in \{1, \dots, m\}$
- For any two consecutive letters g_i and g_{i+1} , they are not in the same group

The motivation for reduced words is to allow us to work with less amount of words since every word can be reduced by removing the identities and replacing two consecutive elements in the same group with their product in that group.

Proposition 1.2.3

Let $\{G_i \mid i \in I\}$ be a collection of groups. Recursively define a product on the set of reduced words as follows.

$$(g_1 \cdots g_m) * (h_1 \cdots h_n) = \begin{cases} g_1 \cdots g_m h_1 \cdots h_n & \text{if } g_m, h_1 \text{ are not in the same group} \\ g \cdots g_{m-1} * g_{m+1} * h_2 \cdots h_n & \text{if } g_m, h_1 \in G_i \text{ for some } i \in I \\ & \text{and } g_m h_1 = g_{m+1} \neq 1 \\ g_1 \cdots g_{m-1} * h_2, \dots, h_n & \text{if } g_m h_1 \in G_i \text{ for some } i \in I \text{ and } g_m h_1 = 1 \end{cases}$$

Then F_S is a group with the above operation.

Proof

We clearly have the identity 1 and the inverse of $g = g_1 \cdots g_n$ to be $g_n^{-1} \cdots g_1^{-1}$. We just have to check associativity and closure. By definition of the free product, the free product of reduced words is again a reduced word by definition and finiteness of the length of a reduced word.

For any $g \in G_i$, define the action of left multiplication by $L_g(h) = g * h$ for any $h \in *_i G_i$. We clearly have $L_{g_1} \circ L_{g_2} = L_{g_1 g_2}$ and $L_{g^{-1}} = L_g^{-1}$ and thus $L_g \in \text{Sym}(*_i G_i)$. For any word $g = g_1 \cdots g_m$, the

map $L : *_i G_i \rightarrow \text{Sym}(*_i G_i)$ is injective since for any $g \in *_i G_i$, we have $L_g(1) = g$ which means that $g \neq h$ implies $L_g \neq L_h$. Thus we have embedded the set into the group $\text{Sym}(*_i G_i)$ so that associativity follows. ■

The motivation is that we want to continue reducing the product of the word so that it can be fully reduced again.

Definition 1.2.4 (The Free Product of Groups)

Let $\{G_i \mid i \in I\}$ be a collection of groups. Define the free product of groups to be the set

$$*_i G_i = \{w \mid w \text{ is a reduced word of } \{G_i \mid i \in I\}\}$$

together with the above product $*$.

Note: If $\{G_i\}$ is just a collection of two groups then we write the free product as $G_1 * G_2$.

Proposition 1.2.5

Let $\{G_i \mid i \in I\}$ be a collection of groups. Then $G_k \leq *_i G_i$ for any $k \in I$.

Proof Clearly every element of G_i is in $*_i G_i$ since we can just treat each element as a word. ■

Proposition 1.2.6 (Universal Property of The Free Product)

Let $\{G_i \mid i \in I\}$ be a collection of groups. Then the free product $*_i G_i$ and the inclusion maps $\iota_k : G_k \rightarrow *_i G_i$ satisfies the following universal property. For any collection of group homomorphisms $\{\phi_i : G_i \rightarrow H \mid i \in I\}$, there exists a unique map

$$*_i \phi_i : *_i G_i \rightarrow H$$

and such that the following diagram commutes:

$$\begin{array}{ccc} G_k & \xrightarrow{\lambda_k} & *_i G_i \\ & \searrow \phi_k & \downarrow \exists! *_i \phi_i \\ & & H \end{array}$$

Moreover, $*_i G_i$ is the unique group (up to unique isomorphism) that has such a property.

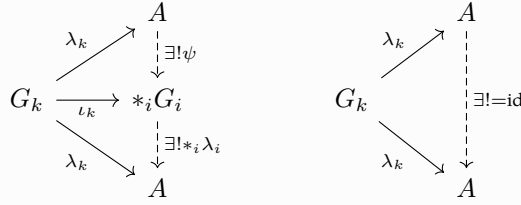
Proof

Define $*_i \phi_i : *_i G_i \rightarrow H$ by $*_i \phi_i(g_1 \cdots g_m) = \phi_{i_1}(g_1) \cdots \phi_{i_m}(g_m)$ where we assume that each $g_k \in G_{i_k}$. Then clearly we have that $(*_i \phi_i) \circ \iota_i = \phi_i$. Moreover, since ϕ_i are group homomorphisms, we have that $\phi_i(g_k) \phi_i(g_{k+1}) = \phi_i(g_k g_{k+1})$ and $\phi_i(1) = 1$ and thus ϕ_i is compatible with reduced words. This means that $*_i \phi_i$ is a group homomorphism and thus proving the existence part of the statement.

Now suppose that A is such a group that satisfies the above property. Consider the following two diagrams:

$$\begin{array}{ccc} & *_i G_i & \\ \iota_k \nearrow & & \downarrow \exists! *_i \lambda_i \\ G_k & \xrightarrow{\lambda_k} & A \\ \iota_k \searrow & & \downarrow \exists! \psi \\ & *_i G_i & \end{array} \quad \begin{array}{ccc} & *_i G_i & \\ \iota_k \nearrow & & \downarrow \exists! = \text{id} \\ G_k & \xrightarrow{\lambda_k} & A \\ \iota_k \searrow & & \downarrow \exists! \psi \\ & *_i G_i & \end{array}$$

Since all the constructed maps are unique such that the diagram commutes, we conclude that $\psi \circ *_i \lambda_i = \text{id}$. Similarly, by considering the following diagrams:



We conclude that $*_i \lambda_i \circ \psi = \text{id}$. Hence $*_i G_i$ and A are isomorphic. Moreover, the isomorphism is unique since all maps constructed are unique such that their respective diagrams commute. ■

1.3 Normal Closure

Definition 1.3.1 (Normal Closure)

Let G be a group. Let $S \subseteq G$ be a subset of G . Define the normal closure of S in G to be the subgroup

$$\langle S^G \rangle = \langle g^{-1}sg \mid \text{for all } s \in S \text{ and } g \in G \rangle$$

Lemma 1.3.2

Let G be a group. Let $S \subseteq G$ be a subset of G . Then $\langle S^G \rangle$ is a normal subgroup of G .

Lemma 1.3.3 Let G be a group. Let $S \subseteq G$ be a subset of G . Then

$$\langle S^G \rangle = \bigcap_{S \subseteq N \trianglelefteq G} N$$

Lemma 1.3.4 Let G be a group. Let $N \trianglelefteq G$ be a normal subgroup of G . Then $\langle N^G \rangle = N$.

Definition 1.3.5 (Normally Generated Groups) Let G be a group. We say that G is normally generated if there exists a subset $S \subseteq G$ such that

$$G = \langle S^G \rangle$$

1.4 Presentations of a Group

Definition 1.4.1 (Group of Generators and Relations) Let S be a set. Let F_S be the free group on S . Let R be a set of words on S . Define the group with generators S and relations R to be the quotient

$$\langle S \mid R \rangle = \frac{F_S}{\langle R^{F_S} \rangle}$$

Elements of S are called generators and elements of R are called relations.

Proposition 1.4.2 (Universal Property of the Group of Generators and Relations) Let S be a set. Let R be a set of words on S . Then the group $\langle S \mid R \rangle$ and the inclusion map $\iota : S \hookrightarrow \langle S \mid R \rangle$ satisfies the following universal property.

For any group G and any map of sets $f : S \rightarrow G$ for which the induced map $\bar{f} : F_S \rightarrow G$ by the universal property of free groups is such that $\bar{f}(r) = 1_G$ for all $r \in R$, there exists a unique group homomorphism $\phi : \langle S \mid R \rangle \rightarrow G$ such that the following diagram commutes:

$$\begin{array}{ccc} S & \xrightarrow{\iota} & \langle S \mid R \rangle \\ & \searrow f & \downarrow \exists! \phi \\ & & G \end{array}$$

Moreover, $\langle S \mid R \rangle$ is the unique group (up to unique isomorphism) that has such a property.

Definition 1.4.3 (Presentations of a Group) Let G be a group. A presentation of G is a group isomorphism

$$G \cong \langle S \mid R \rangle$$

for some set S and some set of words R of S .

Proposition 1.4.4 Every group has a presentation.

Proof This follows from the fact that every group is the quotient of a free group. ■

Example 1.4.5

Let $n \geq 3$. Then there is a group isomorphism

$$D_{2n} \cong \langle r, s \mid r^n, s^2, rsrs \rangle$$

Definition 1.4.6 (Finitely Presented Groups) Let G be a group. We say that G is finitely presented if there exists a presentation

$$G \cong \langle S \mid R \rangle$$

of G such that S and R are finite sets.

1.5 Amalgamated Products

Definition 1.5.1 (Amalgamated Products) Let G, H, K be groups. Let $\phi : K \rightarrow G$ and $\psi : K \rightarrow H$ be injective group homomorphisms. Define the amalgamated product of G and H over K to be the quotient of the free product

$$G *_K H = \frac{G * H}{\langle \{\phi(k)(\psi(k))^{-1} \mid k \in K\}^G \rangle}$$

Example 1.5.2

Let G, H be groups. Then $G *_K H = G * H$.

Lemma 1.5.3

Let G, H, K be groups. Let $\phi : K \rightarrow G$ and $\psi : K \rightarrow H$ be injective group homomorphisms. Then G, H are subgroups of $G *_K H$.

Proposition 1.5.4 (Universal Property of The Free Product)

Let G, H, K be groups. Let $\phi : K \rightarrow G$ and $\psi : K \rightarrow H$ be injective group homomorphisms. Then the amalgamated product $G *_K H$ satisfies the following universal property. For group A and any group homomorphism $p : G \rightarrow A$ and $q : H \rightarrow A$ such that $p \circ \phi = q \circ \psi$, there exists a unique homomorphism

$$G *_K H \rightarrow A$$

such that the following diagram commutes:

$$\begin{array}{ccccc}
 K & \xrightarrow{\phi} & G & & \\
 \psi \downarrow & & \downarrow \text{incl} & \searrow p & \\
 H & \xrightarrow{\text{incl}} & G *_K H & \xrightarrow{\exists!} & A \\
 & \searrow q & & &
 \end{array}$$

Moreover, $G *_K H$ is the unique group (up to unique isomorphism) that has such a property.

1.6 Finitely Generated Groups

Definition 1.6.1 (Finitely Generated Groups) A group G is finitely generated if there is a finite subset $A = \{g_1, \dots, g_n\}$ of G such that $G = \langle A \rangle$.

Proposition 1.6.2 Let G be a finitely generated group. Let N be a normal subgroup of G . Then G/N is finitely generated.

Proposition 1.6.3 Let G be a group. Then G is finitely generated if and only if there exists $n \in \mathbb{N}$ such that G is isomorphic to the quotient of the free group F_n of rank n with a normal subgroup N of F_n , written as

$$G \cong \frac{F_n}{N}$$

Proposition 1.6.4 Let G be a group. If G is finitely presented, then G is finitely generated.

The converse is in general not true. Examples: Higgs group, Lamplighter group.

2 Free Abelian Groups

2.1 Direct Sums of Abelian Groups

Definition 2.1.1 (Direct Sum of Abelian Groups) Let $\{G_i \mid i \in I\}$ be a collection of abelian group. Define the direct sum of the abelian groups to be the set

$$\bigoplus_{i \in I} G_i = \{(g_i)_{i \in I} \in \prod_{i \in I} G_i \mid \text{Only finitely many } g_i \text{ is non-zero}\}$$

together with the product of their binary operations.

Lemma 2.1.2 Let $\{G_i \mid i \in I\}$ be a collection of abelian group. Then $\bigoplus_{i \in I} G_i$ is a subgroup of $\prod_{i \in I} G_i$.

Lemma 2.1.3 Let $G_1, \dots, G_n$ be a finite collection of abelian groups. Then the identity map of sets give a group isomorphism

$$\bigoplus_{i=1}^n G_i \cong \prod_{i=1}^n G_i$$

2.2 Free Abelian Groups

Definition 2.2.1 (Free Abelian Group On a Set) Let S be a set. Define the free abelian group on S to be the direct sum

$$\mathbb{Z}S = \mathbb{Z}\langle s \in S \rangle = \bigoplus_{s \in S} \mathbb{Z}$$

Proposition 2.2.2 (Universal Property of Free Abelian Groups) Let S be a set. Then the free group $\mathbb{Z}S$ and the inclusion map $\iota : S \rightarrow \mathbb{Z}S$ satisfies the following universal property. For any abelian group A and any map of sets $\psi : S \rightarrow A$, there exists a unique group homomorphism $\phi : \mathbb{Z}S \rightarrow A$ such that the following diagram commutes:

$$\begin{array}{ccc} S & \xrightarrow{\iota} & \mathbb{Z}S \\ & \searrow \psi & \downarrow \exists! \phi \\ & & A \end{array}$$

Moreover, $\mathbb{Z}S$ is the unique group (up to unique isomorphism) that has such a property.

Proposition 2.2.3 Let S be a set. Then there is an isomorphism

$$\mathbb{Z}S \cong (F_S)^{\text{ab}}$$

given by the universal property of free abelian groups.

Proposition 2.2.4 Let S, T be sets. Then $\mathbb{Z}S \cong \mathbb{Z}T$ are isomorphic if and only if $|S| = |T|$.

Proposition 2.2.5 Let A be an abelian group and B a subgroup of A . If A/B is a free abelian group, then

$$A \cong B \times A/B$$

Proof Denote the quotient map $f : A \rightarrow A/B$. Since A/B is free abelian, there exists $K = \{a_1 + B, \dots, a_n + B, \dots\}$ such that K is a basis of A/B . Now consider $L = \langle a_1, \dots, a_n, \dots \rangle$. I claim that this is isomorphic to A/B . In fact, this is given by $f|_L$. Clearly $f|_K(a_i) = a_i + B$ for

each i . Suppose that $a \in \ker(f|_K)$. Write $a = \sum_{i=1}^n k_i a_i$. Then we have

$$\begin{aligned} f|_L \left(\sum_{i=1}^n k_i a_i \right) &= 0 \\ \sum_{i=1}^n k_i f|_L(a_i) &= 0 \\ \sum_{i=1}^n k_i (a_i + B) &= 0 \end{aligned}$$

By property of the basis, we have that $k_1 = \dots = k_n = 0$. This generalizes well to the countable case since every element is written in a finite sum. Thus we have that $\ker(f|_L) = 0$. For surjectivity, if $\sum k_i (a_i + B)$ is a finite sum in A/B then just choose $\sum k_i a_i \in L$. Thus we have shown bijectivity. Finally, we show that $A \cong B \times L$. Define $\phi : B \times L \rightarrow A$ by

$$\phi \left(b, \sum k_i a_i \right) = \sum k_i a_i + b$$

We first show injectivity. We have that

$$\begin{aligned} \sum k_i a_i + b &= 0 \\ \sum k_i a_i &= -b \\ \sum k_i f|_L(a_i) &= 0 \\ \sum k_i (a_i + B) &= 0 \end{aligned}$$

This implies that $k_i = 0$ and thus $b = 0$. Thus $\ker(\phi) = 0$. Now suppose that $a \in A$. Consider the element $a + B \in A/B$. Since A/B has the basis K , we have that

$$a + B = \sum k_i (a_i + B) = \left(\sum k_i a_i \right) + B$$

This implies that $a - \sum k_i a_i \in B$. Then choosing $b = a - \sum k_i a_i$ and $\sum k_i a_i \in L$, we are done with surjectivity. This makes sense even if $\sum k_i a_i \in B$ because in this case, $\sum k_i a_i = 0$ and we can choose $b = a \in B$. ■

Given an arbitrary abelian group, we can recognize whether it is free abelian using the concept of a basis that is analogous to that in Linear Algebra.

Definition 2.2.6 (Basis of an Abelian Group) Let G be an abelian group. Let $B \subset G$.

- We say that elements of B are linearly independent if $\sum_{b \in B} n_b \cdot b = 0$ implies $n_b = 0$ for each $b \in B$
- We say that B is a basis of G if it is linearly independent and generate G .

Proposition 2.2.7 Let G be an abelian group. Then G is isomorphic to a free abelian group if and only if G has a basis.

3 Finite and Finitely Generated Abelian Groups

3.1 Finite Abelian Groups

Definition 3.1.1 (Finite Abelian Groups) A group G is said to be a finite abelian group if it is finite and abelian.

Lemma 3.1.2 Let G be a finite abelian group of order n . If p is a prime that divides n , then G contains an element of order p .

Lemma 3.1.3 A finite abelian group is a p group if and only if the order of every element is a power of p .

Lemma 3.1.4 Let G be a finite abelian group. Let H be a subgroup of G . Then there exists a complement K such that $G = H \times K$.

Lemma 3.1.5 Let G be a finite abelian p group and suppose that $g \in G$ is an element in G with the highest order. Then G is isomorphic to $\langle g \rangle \times H$ for some subgroup H of G .

Theorem 3.1.6 (The Fundamental Theorem for Finite Abelian Group (Primary Decomposition)) Let G be a finite abelian group. Then G can be decomposed into a direct product of cyclic groups

$$G \cong \mathbb{Z}/k_1\mathbb{Z} \times \cdots \times \mathbb{Z}/k_n\mathbb{Z}$$

where $k_1, \dots, k_n \in \mathbb{N}$ are such that $k_{i+1} | k_i$ for $1 \leq i \leq n-1$.

Theorem 3.1.7 (Invariant Factor Decomposition) Let G be a finite abelian group. Then G can be decomposed into a direct product of cyclic groups

$$G \cong \mathbb{Z}/k_1\mathbb{Z} \times \cdots \times \mathbb{Z}/k_n\mathbb{Z}$$

where $k_1, \dots, k_n \in \mathbb{N}$ are such that $k_1, \dots, k_n$ are powers of primes that are not necessarily distinct.

3.2 Finitely Generated Abelian Groups

Definition 3.2.1 (Finitely Generated Abelian Groups) A group G is said to be finitely generated abelian if it is finitely generated and abelian.

Lemma 3.2.2 Let G be an abelian group. If G is finite, then G is finitely generated.

Proof A finite choice of generators for G in this case would be the set G . ■

Contrasting finitely generated groups, finitely generated abelian groups allow for a more simple description of subgroups generated by a subset because we can collect like terms by commuting elements. In particular, if G is an abelian group and $A \subseteq G$ is a subset, then

$$\langle A \rangle = \left\{ \sum_{g \in A} k_g \cdot g \mid k \in \mathbb{Z} \right\}$$

Proposition 3.2.3 Subgroups of a finitely generated abelian group is finitely generated.

Proof Let $H \leq G$. We prove by induction on n that H can be generated by at most n elements. If $n = 1$, then G is cyclic. Then $G = \{kx \mid k \in \mathbb{Z}\}$. Let m be the smallest positive number such that $mx \in H$. If m does not exist then $K = \{0\}$ and we are done. Otherwise, $\{k(mx) \mid k \in \mathbb{Z}\} \subseteq H$.

We now prove the reverse inclusion. Write $tx = (qm + r)x \in G$ with $0 \leq r < m$. Then $tx \in H$ if and only if $rx = (t - qm)x \in H$. But this is true if and only if $r = 0$ since m is already the smallest number such that $mx \in H$. Thus $H \subseteq \{k(mx) | k \in \mathbb{Z}\}$ and H is cyclic.

Now suppose that the induction hypothesis is true for $n - 1$. Let K be the subgroup generated by $x_1, \dots, x_{n-1}$. By induction, $H \cap K$ is generated by $y_1, \dots, y_{m-1}$ with $m \leq n$. If $H \leq K$, then $H = H \cap K$ and we are done.

Suppose that H is not a subgroup of K . Then there exists elements of the form $k + tx_n \in H$ with $k \in K$ and $t > 0$. Choose $y_m = h + tx_n$ with t minimal. I claim that H is generated by $y_1, \dots, y_m$. Let $h \in H$. Then $h = k' + ux_n$ with $k' \in K$ and $u \in \mathbb{Z}$. If t does not divide u then we can write $u = tq + r$ with $q, r \in \mathbb{Z}$ and $0, r < t$. Then $k - qy_m = (h' - qh) + rx_n \in H$ thus contradicting minimality of t . Thus $t|u$ and hence $u = tq$ and $k - qy_m \in H \cap K$. But $H \cap K$ is generated by $y_1, \dots, y_{m-1}$ thus we are done. ■

Proposition 3.2.4 Let G be a finitely generated abelian group generated by the elements $g_1, \dots, g_n \in G$. Then there exists a surjective group homomorphism $\phi : \mathbb{Z}^n \rightarrow G$ given by sending $(0, \dots, 0, 1, 0, \dots, 0)$ with non-zero entry at the k th column to g_k .

A free abelian group is in some sense freely generated by its set of basis whose only restriction is for it to be abelian. On the other hand, finitely generated abelian groups may have extra relations between basis elements preventing them from being free. Therefore every finitely generated abelian group can be thought of as a free abelian group, then quotient some non-trivial relations.

Proposition 3.2.5 Let G be a finitely generated abelian group generated by n elements. Then the following are true.

- There exists a finitely generated subgroup $H = \langle v_1, \dots, v_m \rangle$ of $\mathbb{Z}^n$ such that there is an isomorphism

$$G \cong \frac{\mathbb{Z}^n}{H}$$

- G has the presentation of $G \cong \langle x_1, \dots, x_n \mid v_1, \dots, v_m \rangle$ where $x_1, \dots, x_n$ are the standard basis of $\mathbb{Z}^n$ (as a subgroup of $\mathbb{Z}^n$ so that the commuting relations are hidden)

Proof Let G be a finitely generated abelian group. Then there exists $g_1, \dots, g_n \in G$ such that $G = \langle g_1, \dots, g_n \rangle$. Define a function $\phi : \mathbb{Z}^n \rightarrow G$ by sending $(0, \dots, 0, 1, 0, \dots, 0)$ where the k th position is non-zero, to the element g_k . By the universal property of free abelian groups, ■

We can compare some of the similarities between free groups and free abelian groups.

Free Groups	Free Abelian Groups
Groups with the least relations	Abelian groups with the least relations
Unique up to cardinality of underlying set	Unique up to cardinality of basis (which is its underlying set)
Free product of copies of $\mathbb{Z}$	Direct sum of copies of $\mathbb{Z}$
It is finitely generated if and only if it is isomorphic to F_n	It is finitely generated if and only if it is isomorphic to $\mathbb{Z}^n$
Every finitely generated group is isomorphic to a quotient of a free group of finite rank	Every finitely generated abelian group is isomorphic to a quotient of a free abelian group of finite rank

A more surprising result would be the following, which says that if there are no elements in the finitely generated group G with finite order, then it means that no extra relations are imposed on G so that G is a free abelian group.

Proposition 3.2.6 Let G be a finitely generated abelian group. Then G is a free abelian group if and only if G is torsion-free.

In particular, this representation means that $v_1 = 0, \dots, v_m = 0$ in $\mathbb{Z}^n$. The question remaining is to how

to transform the kernel $\ker(\phi)$ into the generators $v_1, \dots, v_m \in \mathbb{Z}^n$.

Proposition 3.2.7 Let G be a finitely generated abelian group. Then there exists a basis $\{b_1, \dots, b_n\}$ of $\mathbb{Z}^n$ such that there is an isomorphism

$$G \cong \langle b_1, \dots, b_n \mid d_1 b_1, \dots, d_m b_m \rangle$$

where $d_1, \dots, d_m \in \mathbb{N} \setminus \{0\}$ are such that $d_i \mid d_{i+1}$ for $1 \leq i \leq m-1$.

Proof Suppose that $G = \langle x_1, \dots, x_n \mid v_1, \dots, v_m \rangle = \frac{\mathbb{Z}^n}{K}$. Since K is a subgroup of $\mathbb{Z}^n$, we know that it is also a free group. ■

Theorem 3.2.8 (Fundamental Theorem of Finitely Generated Abelian Groups (Invariant Factor Decomposition)) Let G be a finitely generated abelian group. Then G can be decomposed into

$$G \cong \mathbb{Z}/n_1\mathbb{Z} \times \cdots \times \mathbb{Z}/n_s\mathbb{Z} \times \mathbb{Z}^r$$

where

- (free rank / Betti number) $r \geq 0$ and $n_j \geq 2$ for all j
- (invariant factors) $n_{i+1} \mid n_i$ for $1 \leq i \leq s-1$
- $|G| = n_1 \cdots n_s$

This expression is unique up to reordering of the external product.

Proof Let G be a finitely generated abelian group. Denote the torsion subgroup of G by tG . Then G/tG is a torsion free abelian group and thus is free abelian. Thus we have that $G \cong tG \times G/tG$. Since tG is a finite abelian group, we can invoke the fundamental theorem for finite abelian groups to get $tG \cong \mathbb{Z}/k_1\mathbb{Z} \times \cdots \times \mathbb{Z}/k_s\mathbb{Z}$. Since G/tG is free abelian, it is isomorphic to $\mathbb{Z}^r$ for some r . Combining the results gives our proof. ■

Below is the Primary Decomposition of finite abelian groups.

Theorem 3.2.9 (Primary Decomposition) Let G be a finitely generated abelian group. Then G can be decomposed into

$$G \cong \mathbb{Z}/q_1\mathbb{Z} \times \cdots \times \mathbb{Z}/q_t\mathbb{Z} \times \mathbb{Z}^r$$

where

- $r \geq 0$ is the free rank of G
- $q_1, \dots, q_t$ are powers of primes numbers that may not be unique.

This expression is unique up to reordering of the indices.

Theorem 3.2.10 Let G be a finitely generated abelian group represented by

$$\langle b_1, \dots, b_n \mid d_1 b_1, \dots, d_m b_m \rangle$$

Further suppose that G has invariant decomposition

$$\left(\bigoplus_{i=1}^m \mathbb{Z}/d'_i\mathbb{Z} \right) \oplus \mathbb{Z}^{n'-m'}$$

Then these two representations of G match up through their numbers. In particular,

- Up to reordering, we have $d_i = d'_i$
- $n = n'$ and $m = m'$

3.3 Unimodular Smith Normal Form

Definition 3.3.1 (Unimodular Elementary Operations) We define unimodular elementary operations on $A \in M_{m \times n}(\mathbb{Z})$ as follows:

- (UR1): Replace row r_i with $r_i + tr_j$ where $j \neq i$ and $t \in \mathbb{Z}$
- (UR2): Interchange two rows r_i and r_j
- (UR3): Replace row r_i by $-r_i$
- (UC1): Replace column c_i with $c_i + tc_j$ where $j \neq i$ and $t \in \mathbb{Z}$
- (UC2): Interchange two columns c_i and c_j
- (UC3): Replace columns r_i by $-r_i$

Definition 3.3.2 (Unimodular Elementary Matrices) Define three elementary matrices as follows:

- Recombine Matrix: The $n \times n$ recombine matrix $R_{i,j,a}$ is given the zero matrix except the diagonal is all 1 and the i th row and j th column has the value $a \in \mathbb{Z}$.
- Scale Matrix: The $n \times n$ scale matrix $R_i(-1)$ is given by the zero matrix except the diagonal is all 1, and the i , i th element is -1 .
- Transposition Matrix: The $n \times n$ transposition matrix $R_{i,j}$ is given by the zero matrix except the diagonal is all 1, the i , i th element and j , j th element is 0 and the i , j th and j , i th element is 1

Proposition 3.3.3 The unimodular elementary operations are exactly equivalent to performing left and right matrix multiplications with unimodular elementary matrices. In particular, left multiplication corresponds to row operations and right multiplication corresponds to column operations.

Theorem 3.3.4 Let $A \in \mathbb{Z}_{m \times n}$ with rank r . Then there exists a sequence of unimodular elementary row and column operations such that A is reduced to

$$\begin{pmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & 0 \\ 0 & 0 & \cdots & 0 \end{pmatrix}$$

where the d_r is the last non zero diagonal entry. The diagonal also satisfies $d_i > 0$ for $1 \leq i \leq r$ and $d_i | d_{i+1}$ for $1 \leq i < r$. The $d_1, \dots, d_r$ are also uniquely determined by A .

Proof ■

Definition 3.3.5 (Unimodular Smith Normal Form) Let $A \in M_{n \times n}(\mathbb{Z})$ be a matrix with coefficients in $\mathbb{Z}$. Define the smith normal form of A to be the unique matrix above, denoted $\text{SNF}(A)$.

Lemma 3.3.6 Let $A \in \mathbb{Z}_{m \times n}$ has unimodular smith normal form S with rank r . Then the greatest common divisor of all entries of A is equal to d_1 where we require $\gcd(r, 0) = r$ for $r \geq 1$.

4 More Types of Groups

4.1 Torsion and Torsion-Free Groups

Definition 4.1.1 (Torsion Elements)

Let G be a group. Let $g \in G$. We say that g is a torsion element if $|g|$ is finite.

Definition 4.1.2 (Torsion Group)

Let G be a group. We say that G is a torsion group if all elements of G are torsion.

Example 4.1.3

Any finite group is torsion.

Definition 4.1.4 (Torsion Free Group)

Let G be a group. We say that G is torsion free if the only torsion element is 1_G .

Definition 4.1.5 (Torsion Subgroup)

Let G be an abelian group. Define the torsion subgroup of G to be

$$G^{\text{tor}} = \{g \in G \mid g \text{ is torsion} \}$$

If G is not abelian, then G^{tor} is not a group.

Lemma 4.1.6

Let G be an abelian group. Then the following are true.

- G^{tor} is a torsion group.
- G/G^{tor} is a torsion free group.

4.2 p-groups

Definition 4.2.1 (p -groups)

Let G be a group. Let p be a prime. We say that G is a p -group if for all $g \in G$, we have $|g| = p^n$ for some $n \in \mathbb{N}$.

Lemma 4.2.2

Let G be a finite group. Let p be a prime. Then G is a p -group if and only if $|G| = p^n$ for some $n \in \mathbb{N}$.

Proposition 4.2.3

Let p be a prime. Let G be a non-trivial finite p -group. Then $|Z(G)| > 1$.

Proposition 4.2.4

Let p be a prime. Let G be a finite p -group of order $|G| = p^n$ for some $n \in \mathbb{N}$. Then for each $0 \leq k \leq n$, there exists a normal subgroup $N \trianglelefteq G$ such that $|N| = p^k$.

4.3 Simple Groups

Definition 4.3.1 (Simple Groups)

Let G be a group. We say that G is simple if $\{1_G\}$ and G are the only normal subgroups of G .

Proposition 4.3.2

Let G be an abelian group. Then G is simple if and only if $G \cong C_p$ for some prime p .

4.4 Abelianization and Commutators

Commutators are useful for checking whether a given group is cyclic or not.

Definition 4.4.1 (Commutators) Let $g, h \in G$. Define the commutator of g and h to be

$$[g, h] = ghg^{-1}h^{-1}$$

Definition 4.4.2 (Commutator Subgroup) Let G be a group. Define the commutator subgroup of G to be

$$[G, G] = \langle [g, h] \mid g, h \in G \rangle$$

Note that in general, elements of $[G, G]$ are products of several commutators because we are taking the group generated by all the commutators. In general, the set of all commutators $\{[g, h] \mid g \in G, h \in H\}$ is not a group.

Example 4.4.3

Let $n \geq 3$. Suppose that a presentation of D_{2n} is given by $D_{2n} = \langle r, s \mid r^n, s^2, rsrs \rangle$. Then we have $[D_{2n}, D_{2n}] = \langle r^2 \rangle$.

Proof

A general element of D_{2n} under the above presentation is of the form $s^i r^j$ for $0 \leq i \leq 1$ and $0 \leq j \leq n-1$. We compute that $[s^{i_1} r^{j_1}, s^{i_2} r^{j_2}] = r^{2(j_1 - j_2)}$. Thus the commutator of D_{2n} is given by $\langle r^2 \rangle$. ■

Proposition 4.4.4 Let G be a group. Then the following hold for the commutator subgroup of G .

- $[G, G] \trianglelefteq G$
- $G/[G, G]$ is abelian
- If $N \trianglelefteq G$ and G/N are abelian, then $[G, G] \leq N$.

Proof

- Let $g, h, k \in G$. Then we have

$$\begin{aligned} k[g, h]k^{-1} &= kghg^{-1}h^{-1}k^{-1} \\ &= (kgk^{-1})(khk^{-1})(kg^{-1}k^{-1})(kh^{-1}k^{-1}) \\ &= [kgk^{-1}, khk^{-1}] \in [G, G] \end{aligned}$$

So for $[g_1, h_1], \dots, [g_n, h_n] \in [G, G]$, we have

$$k[g_1, h_1] \cdots [g_n, h_n]k^{-1} = (k[g_1, h_1]k^{-1}) \cdots (k[g_n, h_n]k^{-1}) \in [G, G]$$

- Let $g, h \in G$. Then we have

$$\begin{aligned} ghg^{-1}h^{-1} \in [G, G] &\implies [G, G]ghg^{-1}h^{-1} = [G, G] \\ &\implies [G, G]gh = [G, G]hg \\ &\implies [G, G]g \cdot [G, G]h = [G, G]h \cdot [G, G]g \end{aligned}$$

So $G/[G, G]$ is abelian.

- Suppose that G/N is abelian. Then $Ng \cdot Nh = Nh \cdot Ng$ for all $g, h \in G$. Thus

$$N(ghg^{-1}h^{-1}) = N$$

and so $[g, h] \in N$. Since N is a subgroup of G , we must have that $[G, G] \leq N$.

And so we conclude. ■

Definition 4.4.5 (Abelianization) Let G be a group. Define the abelianization of G to be the abelian group

$$G^{\text{ab}} = \frac{G}{[G, G]}$$

Example 4.4.6

Let $n \geq 3$. Then the following are true.

- If n is odd, then $(D_{2n})^{\text{ab}} \cong C_2$.
- If n is even, then $(D_{2n})^{\text{ab}} \cong C_2 \times C_2$.

Proof

Recall we computed that $[D_{2n}, D_{2n}] = \langle r^2 \rangle$. If n is odd, then $\langle r^2 \rangle = \langle r \rangle$ and $|(D_{2n})^{\text{ab}}| = 2$. The only possible group for the abelianization is C_2 in this case.

If n is even, consider the homomorphism $D_{2n} \rightarrow C_2 \times C_2$ given by $r \mapsto (1, 0)$ and $s \mapsto (0, 1)$. This is well defined by the universal property of the group of generators and relations. The kernel of this map is precisely $\langle r^2 \rangle$. By the first isomorphism theorem, we have a map $(D_{2n})^{\text{ab}} \rightarrow C_2 \times C_2$ that is surjective. Hence it is an isomorphism. ■

Proposition 4.4.7

A group G is abelian if and only if $G = G^{\text{ab}}$.

Proof It suffices to show that G is abelian if and only if $[G, G] = 1$. We know that if G is abelian, then every commutator reduces to

$$[g, h] = ghg^{-1}h^{-1} = gh h^{-1}g^{-1} = 1$$

so $[G, G] = 1$. Now suppose that $[G, G] = 1$. Then this implies that $[g, h] = ghg^{-1}h^{-1} = 1$ which shows that $gh = hg$ for any $g, h \in G$. Thus G is abelian. ■

Proposition 4.4.8 (Universal Property of Abelianization)

Let G be a group. Then the abelianization G^{ab} of G together with the quotient homomorphism $G \rightarrow G^{\text{ab}}$ satisfies the following universal property: For any abelian group A and any group homomorphism $\psi : G \rightarrow A$, there exists a unique group homomorphism $\phi : G^{\text{ab}} \rightarrow A$ such that the following diagram commutes:

$$\begin{array}{ccc} G & \xrightarrow{\text{proj}} & G^{\text{ab}} \\ & \searrow \psi & \downarrow \exists! \phi \\ & & A \end{array}$$

Moreover, G^{ab} is the unique group (up to unique isomorphism) that has such a property.

Definition 4.4.9 (Perfect Groups)

Let G be a group. We say that G is perfect if $[G, G] = G$.

Lemma 4.4.10

Let G be a group. If G is simple, then G is perfect.

Proof

Follows from the fact that $[G, G] \trianglelefteq G$ is a normal subgroup and G is simple. ■

5 The Sylow Theorems and its Consequences

Lagrange's theorem leads to a natural question. Does the converse hold? That is, given a number k dividing $|G|$, is there a subgroup of G with order k ? This is true for cyclic groups but in general it is not true. For example, if $|G|$ is a non-abelian finite simple group, then G has no subgroup of order $|G|/2$ because otherwise, this subgroup would be normal.

We then want sufficient criterion for the converse to hold. This leads to the Sylow theorems and its relation to p -groups.

5.1 The Four Sylow Theorems

We introduce a notation for subgroups of order a power of a prime.

Definition 5.1.1 (Sylow p -groups) Let G be a group of order $p^k m$ where $\gcd(p, m) = 1$, then subgroups of G of order p^k is called a Sylow p -subgroup of G . The set of all Sylow p -subgroups of G is denoted

$$\text{Syl}_p(G) = \{H \leq G \mid H \text{ is a Sylow } p\text{-subgroup}\}$$

and the number of Sylow p -subgroups is denoted

$$n_p(G) = |\text{Syl}_p(G)|$$

The maximal power of the prime p that divides $|G|$ is also called the p -part of G . This is denoted as $|G|_p$. Before we prove the main theorems of the section, we need a lemma.

Lemma 5.1.2 Let p be prime. Let $n, m \in \mathbb{N}$ such that $\gcd(m, p) = 1$. Then the following are true.

- p divides $\binom{p}{i}$ for all $1 \leq i \leq p-1$
- $\binom{p^n m}{p^n} \equiv m \pmod{p}$

Proof

- By definition, we have that

$$\binom{p}{i} = \frac{p(p-1) \cdots (p-(i-1))}{i(i-1) \cdots 2 \cdot 1}$$

Let $a = (p-1) \cdots (p-(i-1))$ and $b = i!$ so that $\binom{p}{i} = \frac{pa}{b}$. Since b is the product of integers less than p , all prime divisors of b are less than p . Thus p does not divide a . We have that $b\binom{p}{i} = pa$. Since $\binom{p}{i}$ is an integer, and p does not divide b , p must divide $\binom{p}{i}$.

- Denote $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$. Consider the polynomial ring $\mathbb{F}_p[x]$ and $(1+x)^p \in \mathbb{F}_p[x]$. By the binomial theorem, we have that

$$\begin{aligned} (1+x)^p &= \sum_{i=0}^p \binom{p}{i} x^i \\ &= \binom{p}{0} + \binom{p}{p} x^p \\ &= 1 + x^p \end{aligned}$$

A similar calculation also shows that $(1+x)^{p^n} = 1 + x^{p^n}$. Thus

$$(1+x)^{p^n m} = (1+x^{p^n})^m$$

Using the binomial theorem on both sides, we get

$$\sum_{i=0}^{p^n m} \binom{p^n m}{i} x^i = \sum_{j=0}^m \binom{m}{j} x^{p^n j}$$

Comparing coefficients give

$$\binom{p^n m}{p^n j} \equiv \binom{m}{j} \pmod{p}$$

Applying the result with $j = 1$ proves the lemma.

Thus we are done. ■

The four Sylow theorems is due to Ludwig Sylow. It is an important corner stone in finite group theory.

Theorem 5.1.3 (The First and Second Sylow Theorems) Let G be a finite group and $|G| = p^k m$ where $\gcd(p, m) = 1$. Then the following are true.

- There exists at least one Sylow p -subgroup of G .
- $n_p(G) \equiv 1 \pmod{p}$

Proof We first prove that $n_p(G) \equiv 1 \pmod{p}$. Let G be a finite group and $|G| = p^k m$ where $\gcd(p, m) = 1$. Define

$$X = \{S \subset G \mid |S| = p^n\}$$

Then $|X| = \binom{p^n m}{p^n}$ and thus $|X| \equiv m \pmod{p}$ by the above lemma. Notice that p does not divide $|X|$.

Consider the action of G on X by left multiplication. If $S \in X$, define

$$g \cdot S = gS = \{gs \mid s \in S\}$$

Clearly this is a group action. We know that $\{\text{Orb}_G(S) \mid S \in X\}$ partitions X . In other words, there exists $S_1, \dots, S_r \in X$ such that

$$X = \bigsqcup_{i=1}^r \text{Orb}_G(S_i)$$

and thus $|X| = \sum_{k=1}^r |\text{Orb}_G(S_i)|$.

Since we know that p does not divide $|X|$, there must exist at least one of the orbits with $|\text{Orb}_G(S_i)|$ not divisible by p . Let t be the number of orbits that have order indivisible by p . Without loss of generality, reorder the orbits such that $|\text{Orb}_G(S_i)|$ is not divisible by p for $1 \leq i \leq t$ and $p \mid |\text{Orb}_G(S_i)|$ for $t+1 \leq i \leq r$.

We now prove two claims.

Claim 1: Fix $1 \leq i \leq t$. Then there exists $x_i \in G$ such that $\text{Stab}_G(x_i S_i) = x_i S_i$. In particular, $\text{Stab}_G(x_i S_i) \in \text{Syl}_p(G)$ for each $1 \leq i \leq r$.

Take $1 \leq i \leq t$. Let $s_i \in S_i$ and take $x_i = s_i^{-1}$. Let $T_i = x_i S_i$. Then $1_G \in T_i$. By definition of orbits, we have $\text{Orb}_G(T_i) = \text{Orb}_G(S_i)$. By the orbit stabilizer theorem, we have that p does not divide $|O_{T_i}| = [G : \text{Stab}_G(T_i)]$. Since $|G| = p^n m = [G : \text{Stab}_G(T_i)] |\text{Stab}_G(T_i)|$, we have that $p^n \mid |\text{Stab}_G(T_i)|$. On the other hand, if $g \in \text{Stab}_G(T_i)$, then $gT_i = T_i$ by definition. Since $1_G \in T_i$, we have that $g \in gT_i = T_i$ and thus $\text{Stab}_G(T_i) \subseteq T_i$. Since $|T_i| = p^n \leq |\text{Stab}_G(T_i)|$, we deduce that $\text{Stab}_G(T_i) = T_i$ as required.

Claim 2: $t = |\text{Syl}_p(G)|$.

Define a map $f : \{\text{Orb}_G(T_i) \mid 1 \leq i \leq t\} \rightarrow \text{Syl}_p(G)$ by $f(\text{Orb}_G(T_i)) = T_i$. By claim 1, we know that T_i is a subgroup of G of order p^n since $T_i = \text{Stab}_G(T_i)$, so this map is well defined. We prove that f is a bijection.

Injectivity: Let $1 \leq i, j \leq t$. Suppose that $T_i = T_j$. Then

$$\text{Orb}_G(S_i) = \text{Orb}_G(T_i) = \text{Orb}_G(T_j) = \text{Orb}_G(S_j)$$

so $S_i = S_j$ since we have chosen $S_1, \dots, S_r$ so that their orbits are distinct.

Surjectivity: Let $P \in \text{Syl}_p(G)$. Then $P \in X$, and $\text{Stab}_G(P) = P$. It follows that p does not divide

$m = [G : P] = [G : \text{Stab}_G(P)] = |\text{Orb}_G(P)|$, so $\text{Orb}_G(P) = \text{Orb}_G(T_i)$ for some i . Since $1_G \in T_i$, and $\text{Orb}_G(T_i) = \{xS_i \mid x \in G\}$, T_i is the unique element of $\text{Orb}_G(T_i)$ containing 1_G . Thus we must have $P = T_i = f(\text{Orb}_G(T_i))$ as required.

We resume the proof of the second Sylow theorem. Since we know that $|X| \equiv m \pmod{p}$, and that

$$|X| \equiv \sum_{i=1}^r |\text{Orb}_G(T_i)| \pmod{p}$$

Since $|\text{Orb}_G(T_i)| = [G : \text{Stab}_G(T_i)] = |G|/|T_i| = m$ by claim 1, and $t = |\text{Syl}_p(G)|$ by claim 2. From the above sum we deduce that

$$m \equiv mn_p(G) \pmod{p}$$

Since we also have that $\gcd(m, p) = 1$, we can cancel m on both sides to get $n_p(G) \equiv 1 \pmod{p}$. This also proves the first Sylow theorem. ■

Proposition 5.1.4 Let G be a group and p a prime divisor of $|G|$. Let $H \leq G$ and let P be a Sylow p -subgroup of G . Then there exists some $g \in G$ such that $H \cap gPg^{-1}$ is a Sylow p -subgroup of H .

Proof Consider the set $X = \{xP \mid x \in G\}$ of left cosets of P in G . Then G acts on X by left multiplication and thus H acts on X . For any $xP \in X$, we have that

$$\begin{aligned} \text{Stab}_H(xP) &= \{h \in H \mid hxP = xP\} \\ &= \{h \in H \mid x^{-1}hxP = P\} \\ &= \{h \in H \mid h \in xPx^{-1}\} \\ &= H \cap xPx^{-1} \end{aligned}$$

Since P is a Sylow p -subgroup of G , $|X| = [G : P]$ is coprime to p . Let $\text{Orb}_H(x_1P), \dots, \text{Orb}_H(x_sP)$ denote the distinct orbits of H in its action on X . We know that $|X| = \sum_{i=1}^s |\text{Orb}_H(x_iP)|$. If every such orbit has size divisible by p , then $|X|$ would be divisible by p , which is a contradiction. Thus there exists $x_i \in G$ such that p does not divide $|\text{Orb}_H(x_iP)|$.

We know that $\text{Stab}_H(x_iP) = H \cap x_iPx_i^{-1}$ is a p -subgroup since it is a subgroup of $x_iPx_i^{-1}$. We also have $[H : \text{Stab}_H(x_iP)] = |\text{Orb}_H(x_iP)|$ is not divisible by p . Thus $H \cap x_iPx_i^{-1}$ is a Sylow p -subgroup of H . Thus we are done. ■

Theorem 5.1.5 (The Third and Fourth Sylow Theorems) Let G be a finite group and $|G| = p^k m$ where $\gcd(p, m) = 1$. Then the following are true.

- Every Sylow p -subgroup is conjugate to each other
- Any p -subgroup of G is contained in a Sylow p -subgroup of G

Proof

- Let P and Q be Sylow p -subgroups of G . By the above proposition, there exists $g \in G$ such that $Q \cap gPg^{-1}$ is a Sylow p -subgroup of Q . But $|Q|$ is a power of p and thus $\text{Syl}_p(Q) = \{Q\}$. Thus $Q = Q \cap gPg^{-1}$. Since also $|Q| = |gPg^{-1}|$ we have $Q = gPg^{-1}$.
- Let H be a Sylow p -subgroup of G and P a Sylow p -subgroup of G . By the above proposition, there exists $g \in G$ such that $H \cap gPg^{-1}$ is a Sylow p -subgroup of H . But $|H|$ is a power of p and thus $\text{Syl}_p(H) = \{H\}$. Thus $H = H \cap gPg^{-1} \leq gPg^{-1}$.

Thus we are done. ■

5.2 Consequences of the Sylow Theorems

Corollary 5.2.1 Let G be a finite group. Let p be prime such that $|G| = p^k m$ with $\gcd(p, m) = 1$. Let $P \in \text{Syl}_p(G)$ be a Sylow p -subgroup. Then the following are true with respect to $n_p(G)$.

- $n_p(G) = [G : N_G(P)]$
- $n_p(G) | m$
- $P \trianglelefteq G$ if and only if $n_p(G) = 1$

Proof

- By the third Sylow theorem, we know that G acts on $\text{Syl}_p(G)$ by conjugation and that $\text{Orb}_G(P) = \text{Syl}_p(G)$. The stabilizer $\text{Stab}_G(P)$ is thus equal to the normalizer $N_G(P)$. Using the orbit stabilizer theorem, we have that

$$|\text{Syl}_p(G)| = |\text{Orb}_G(P)| = [G : \text{Stab}_G(P)] = [G : N_G(P)]$$

- We know that P is a subgroup of $N_G(P)$. This means that we can apply Lagrange's theorem to get $|N_G(P)| = |P| \cdot [N_G(P) : P]$. Thus we have that

$$\begin{aligned} |\text{Syl}_p(G)| &= [G : N_G(P)] \\ &= \frac{|G|}{|N_G(P)|} \\ &= \frac{|G|}{|P| \cdot [N_G(P) : P]} \end{aligned}$$

Thus $|\text{Syl}_p(G)|$ divides $\frac{|G|}{|P|} = m$

- We have that P is a normal subgroup of G if and only if $N_G(P) = G$. And this is true if and only if $|\text{Syl}_p(G)| = 1$ by the first item.

Thus we are done. ■

Proposition 5.2.2 There are no simple groups of order 20.

Proof Let G be a group of order 20. Then $|G| = 2^2 \cdot 5$. By Sylow theorem 2, we know that $n_5(G) \equiv 1 \pmod{5}$. By corollary 5.2.1, we have that $n_5(G)$ divides $\frac{|G|}{|P|} = \frac{20}{5} = 4$ and thus $n_5(G) = 1$. By the same corollary, we know that the unique Sylow 5-subgroup is normal. Thus G cannot be simple. ■

Proposition 5.2.3 There are no simple groups of order 48.

Proof Let G be a group of order 48. Then $|G| = 2^4 \cdot 3$. Consider $n_2(G)$. By the second Sylow theorem, we have that $n_2(G) \equiv 1 \pmod{2}$. By corollary 5.2.1, we have that $n_2(G) | 3$ and thus $n_2(G) = 1$ or 3.

Suppose that $n_2(G) = 1$. Then by the same corollary, we have a unique Sylow 2-subgroup that is normal to G . Thus G is not simple.

Suppose that $n_2(G) = 3$. Then G acts on $\text{Syl}_2(G)$ non-trivially. Then we know that there is a homomorphism $\phi : G \rightarrow S_3$. The first isomorphism theorem and Lagrange's theorem tells us that $\frac{|G|}{|\ker(\phi)|} = |\text{im}(\phi)|$. $\text{im}(\phi)$ is a subgroup of S_3 whose order is greater than 1 since ϕ is non-trivial. Thus $1 \leq |\text{im}(\phi)| \leq 6$ and thus $48/6 \leq |\ker(\phi)| < 48$. Thus $\ker(\phi)$ is a non-trivial normal subgroup of G and thus G is not simple. ■

5.3 Simplicity of A_n for $n \geq 5$

Using the Sylow theorems, we can also analyze the number of elements that possess a power of prime order. For a finite group G , define

$$F_p(G) = \{x \in G \mid x \neq 1 \text{ and } |x| = p^n \text{ for some } n\}$$

to be the number of elements in G that has its order the power of a prime p . We can deduce a number of information from the Sylow theorems, so that we can eventually have some sort of contradiction in our proof of simplicity.

Corollary 5.3.1 Let G be a finite group such that $|G| = p^k m$ where p is a prime and $\gcd(p, m) = 1$. Let

$$F_p(G) = \{x \in G \mid x \neq 1 \text{ and } |x| = p^n \text{ for some } n\}$$

Then the following are true.

- $F_p(G) = \bigcup_{P \in \text{Syl}_p(G)} (P \setminus \{1\})$
- $|F_p(G)| \geq p^k - 1$ with equality if and only if $|\text{Syl}_p(G)| = 1$
- If $k = 1$ then $|F_p(G)| = |\text{Syl}_p(G)| \cdot (p - 1)$

Proof

- First let $x \in F_p(G)$. Then $\langle x \rangle$ has a prime power order and so is a p -subgroup of G . Any p -subgroup is contained in a Sylow p -subgroup and so x lies in $\bigcup_{P \in \text{Syl}_p(G)} (P \setminus \{1\})$. Now if y lies in the union of the Sylow p -subgroups, then $\langle y \rangle$ is a subgroup of some Sylow p -subgroup. By Lagrange's theorem, the order of y must be a power of p since any Sylow p -subgroup has prime power order. Thus $y \in F_p(G)$.
- From the above characterization of $F_p(G)$, we have that $F_p(G) \supseteq (P \setminus \{1\})$ for any Sylow p -subgroup. Thus $|F_p(G)| \geq p^k - 1$. Now suppose that equality holds. Suppose for a contradiction that there are two distinct Sylow p -subgroups P and Q . This means that there exists an element in Q not contained in P . Hence $|F_p(G)| \geq |P \setminus \{1\}| + 1 = p^k$, this contradicts the assumption of equality. It is also clear that if $n_p(G) = 1$ then $|F_p(G)| = p^k - 1$.
- Firstly notice that when $k = 1$, we have that the first property in this corollary is a disjoint union. Indeed, if P_i and P_j are Sylow p -subgroups that intersect with element $x \in P_i \cap P_j$, then $\langle x \rangle$ has order p so that $P_i = \langle x \rangle = P_j$.

We will now show that

$$F_p(G) = \coprod_{P \in \text{Syl}_p(G)} (P \setminus \{1\})$$

If $x \in F_p(G)$, then x has order p so that $\langle x \rangle$ is a Sylow p -subgroup and $x \in \langle x \rangle \subseteq \coprod_{P \in \text{Syl}_p(G)} (P \setminus \{1\})$. Now suppose that y is an element that lies in the coproduct. Then y lies in some Sylow p -subgroup. Any Sylow p -subgroup is cyclic and isomorphic to C_p since $k = 1$. Thus y has order p and $y \in F_p(G)$. Now $|F_p(G)| = |\text{Syl}_p(G)| \cdot (p - 1)$ and so we conclude.

This completes the proof. ■

Lemma 5.3.2 Let G be a finite group such that $|G| = p^k m$ where p is a prime and $\gcd(p, m) = 1$. Let

$$F_p(G) = \{x \in G \mid x \neq 1 \text{ and } |x| = p^n \text{ for some } n\}$$

Let N be a normal subgroup of G . Then the following are true.

- If $x \in N$, then $\{gxg^{-1} \mid g \in G\} \subseteq N$
- If p does not divide n , then $\text{Syl}_p(G) = \text{Syl}_p(N)$ and $F_p(G) = F_p(N)$.

Proposition 5.3.3 The alternating group A_5 is simple.

Proof Suppose that there exists N a non-trivial normal subgroup of A_5 . By Lagrange's theorem, $|N|$ divides $|A_5| = 60$ and thus at least one of the prime divisors are either 2, 3 or 5. The 5-cycles of A_5 are precisely elements of A_5 with order 5. Since each 5-cycle is of the form $(a_1 \cdots a_5)$, there are 5 possible choices for a_1 , 4 for a_2 and so on. Since also that moving elements transitively along the cycle gives another presentation of the same cycle, we divide by 5 to obtain that there are precisely 24 elements of order 5 of A_5 . A similar argument show that A_5 has 20 elements of order 3.

The only elements of order 2 are products of two disjoint transpositions. A similar argument shows that there are 15 of them. We know that they are part of the same conjugacy class in A_5 .

Case 1: Either 3 divides $|N|$ or 5 divides $|N|$.

Let $p = 3$ or 5 such that p divides $|N|$. Then we know that p does not divide $[G : N]$ so by the above theorem we deduce that $F_p(G) = F_p(N)$. If $p = 5$ then $F_p(N) = 24$ and thus $|N| \geq 25$. Since $|N| \mid 60$ this implies that $|N| = 30$. If $p = 3$ then $F_p(N) = 20$ and thus $|N| \geq 21$. Since $|N| \mid 60$ we thus have $|N| = 30$. If either 3 divides $|N|$ or 5 divides $|N|$ then both 3 and 5 divides $|N|$. But $F_3(N) + F_5(N) > |N|$, a contradiction.

Case 2: Neither 3 or 5 divides $|N|$.

Then $|N|$ divides 4 by Lagrange's theorem. By Cauchy's theorem, there exists $x \in N$ of order 2. Thus using the above corollary, we have that $4 = |N| \geq \text{Cl}(x) = 15$ which is a contradiction. ■

Lemma 5.3.4 Let $n \geq 3$. Let X be the set of 3-cycles in S_n . Then $A_n = \langle X \rangle$.

Proof By definition, we know that A_n consists of elements of an even number of transpositions. We just have to show that every product of a pair of transpositions can be written as a product of 3-cycles. Let $(a \ b)$ and $(c \ d)$ be a pair of transpositions for $a, b, c, d \in \{1, \dots, n\}$ with $a \neq b$ and $c \neq d$. There are three cases.

Case 1: $\{a, b\} \cap \{c, d\} = \emptyset$.

Then the two cycles are disjoint. By direct calculation, we have that

$$(a \ b)(c \ d) = (a \ b \ c)(b \ c \ d)$$

and so we are done.

Case 2: $|\{a, b\} \cap \{c, d\}| = 1$.

Without loss of generality assume that $a = c$. Then

$$(a \ b)(a \ d) = (a \ d \ b)$$

and so we are done.

Case 3: $|\{a, b\} \cap \{c, d\}| = 2$.

Then $(a \ b)^2 = () = (a, b, e)^3$ for any $e \in \{1, \dots, n\} \setminus \{a, b, c, d\}$ and so we conclude. ■

Lemma 5.3.5 Let $n \geq 5$. Then any two 3-cycles are conjugate in A_n .

Proof We know that gfg^{-1} has the same cycle type as f for $f, g \in S_n$. Let X be the set of 3-cycles in S_n . Then A_n acts on X by conjugation. By the above lemma, the group action is transitive so that it has only one orbit. ■

Lemma 5.3.6 Let $n \geq 5$. Let $\sigma \in A_n$ be a non-trivial element. Then for any conjugate τ of σ , there exists $si \in \{1, \dots, n\}$ such that $\tau(i) = \sigma(i)$.

Proof Let r be the longest length of a disjoint cycle in σ . Relabelling if necessary, we have that

$$\sigma = (1 \ 2 \ \dots \ r) \pi$$

where π and $(1 \ 2 \ \dots \ r)$ are disjoint permutations. If $r \geq 3$. Let $\gamma = (3 \ 4 \ 5)$ and $\tau = \gamma\sigma\gamma^{-1}$. Clearly $\tau \neq \sigma$ since $\sigma(3) = 4$ and $\tau(3) = 6$. But $\sigma(1) = 2 = \tau(1)$ and so we are done.

If $r = 2$, then σ is a product of two disjoint transpositions. If there are at least 3 disjoint transpositions, then $n \geq 6$ and after relabelling we can write $\sigma = (1 \ 2) (3 \ 4) (5 \ 6) \dots$. Let $\gamma = (1 \ 3 \ 2)$ and set $\tau = \gamma\sigma\gamma^{-1}$. Clearly $\tau \neq \sigma$. We also have $\sigma(5) = 5 = \tau(5)$ and so we conclude. ■

Theorem 5.3.7 The alternating group A_n for $n \geq 5$ is simple.

Proof We know that A_5 is simple. So suppose that $n \geq 6$. Recall that A_n acts on $X_n = \{1, \dots, n\}$, inherited by the action of S_n . For each $i \in X_n$, we have that $\text{Stab}_{A_n}(i) \cong A_{n-1}$. We proceed by induction. The case $n = 5$ is clear. So suppose that A_{n-1} is simply. By induction, $\text{Stab}_{A_n}(i)$ is a simple group. Notice that it also contains a 3-cycle.

Assume there exists a non-trivial proper normal group N of A_n . Let $\sigma \in N$ be a non-trivial element of N . By the above lemma, there exists a conjugate $\tau \in A_n$ of σ such that $\tau \neq \sigma$ but $\sigma(i) = \tau(i)$ for some $i \in X_n$. Since $N \trianglelefteq A_n$, $\tau \in N$. Hence $\sigma^{-1}\tau \in N$ and $\sigma^{-1}\tau \neq 1_{A_n}$ and $\sigma^{-1}\tau(i) = i$. Thus $\sigma^{-1}\tau \in \text{Stab}_{A_n}(i)$ and so $N \cap \text{Stab}_{A_n}(i) \neq \{1_{A_n}\}$. N being normal to A_n implies that

$$N \cap \text{Stab}_{A_n}(i) \trianglelefteq \text{Stab}_{A_n}(i)$$

Since $\text{Stab}_{A_n}(i)$ is simple, we have $N \cap \text{Stab}_{A_n}(i) = \text{Stab}_{A_n}(i)$ so that $\text{Stab}_{A_n}(i) \leq N$. But $\text{Stab}_{A_n}(i)$ contains a 3-cycle and thus so does N . By lemma 5.3.4, N contains all 3-cycles of A_n . By lemma 5.3.3, $N = A_n$ which is a contradiction. Thus we conclude. ■

In fact, we can show that A_5 is the unique simple group of order 60. We will prove this once we classified small groups.

6 Some Classification Results of Finite Groups

6.1 Fitting's Lemma

Definition 6.1.1 (Inversion Homomorphism) The inversion homomorphism between two subgroups H and K of G is the homomorphism

$$\phi : H \rightarrow \text{Aut}(K)$$

defined by $\phi_1(k) = k$ and $\phi_h(k) = k^{-1}$ for $h \neq 1$.

Lemma 6.1.2 (Fitting's Lemma) Let G be a finite group. Suppose that $K \trianglelefteq G$ is abelian with odd order $\frac{|G|}{2}$. Let $H = \langle x \rangle \in \text{Syl}_2(G)$. Define

$$[K, x] = \langle [v, x] \mid v \in K \rangle$$

Then the following are true.

- $xax^{-1} = a^{-1}$ for all $a \in [K, x]$
- $K = C_K(x) \times [K, x]$
- $G \cong (H \rtimes_\phi [K, x]) \times C_K(x)$ where $\phi : H \rightarrow \text{Aut}(K)$ is the inversion homomorphism.

Proof

- Since K is normal, $[K, x]$ is contained in K . Moreover, since K is abelian, we have that $xax^{-1} = a^{-1}$ for $a \in [K, x]$ if and only if $xax^{-1} = a^{-1}$ for all a in the generating set of $[K, x]$. So it suffices to prove that $x[v, x]x^{-1} = [v, x]^{-1}$ for all $v \in K$. But we have

$$x(vxv^{-1}x^{-1})x = xvxv^{-1} = [v, x]^{-1}$$

since $x = x^{-1}$ and so we are done.

- Define $f : K \rightarrow [K, x]$ by $f(k) = [k, x]$ for $k \in K$. Then

$$\begin{aligned} f(k_1k_2) &= k_1k_2xk_2^{-1}k_1^{-1}x^{-1} \\ &= k_1k_2xk_2^{-1}x^{-1}xk_1^{-1}x^{-1} \\ &= k_1k_2(xk_2^{-1}x^{-1})(xk_1^{-1}x^{-1}) \\ &= k_1(xk_2^{-1}x^{-1})k_2(xk_1^{-1}x^{-1}) \quad (K \text{ is abelian}) \\ &= [k_1, x][k_2, x] \\ &= f(k_1)f(k_2) \end{aligned}$$

Thus f is a homomorphism. Clearly $\ker(f) = C_K(x)$. Also since $\text{im}(f)$ is a subgroup of $[K, x]$ and contains a generating set for $[K, x]$, we have $\text{im}(f) = [K, x]$. Thus by the first isomorphism theorem, we have that

$$|K| = |\ker(f)||\text{im}(f)| = |C_K(x)||[K, x]|$$

Thus $K = C_K(x)[K, x]$. Since K is abelian, we have that $K = C_K(x) \times [K, x]$ and so we conclude.

- Step 1: $[K, x] \trianglelefteq G$.

Notice that $G = \langle x \cup K \rangle$. We just have to show that every element in the generating set normalizes $[K, x]$. By the first part of this lemma, it is easy to see that x normalizes $[K, x]$. Since K is abelian, K also normalizes $[K, x]$. Thus we are done. ■

6.2 Recognizing Dihedral Groups

With the inversion homomorphism, the following proposition shows that D_{2n} can be decomposed into a semidirect product.

Proposition 6.2.1

Let $\sigma = (1 \ \cdots \ n) \in S_n$ and $\tau = \prod_{i=1}^{\lfloor n/2 \rfloor} (i \ n-i+1) \in S_n$. Write $K = \langle \sigma \rangle$ and $H = \langle \tau \rangle$. Then

$$D_{2n} \cong H \rtimes_{\phi} K$$

where $\phi : H \rightarrow \text{Aut}(K)$ is the inversion homomorphism.

Proof It is clear that $|\tau| = 2$ and $|\sigma| = n$ and $D_{2n} = K \amalg \tau K$. This implies that $G = HK$ and $H \cap K = \{1\}$. Hence $G \cong H \rtimes_{\phi} K$ where $\phi : \{1, \tau\} \rightarrow \text{Aut}(K)$ is given by $\phi_1(k) = k$ and $\phi_{\tau}(k) = k^{-1}$. ■

Theorem 6.2.2

Let G be a nonabelian finite group. Further suppose that

- G has a cyclic subgroup of K of order $n = \frac{|G|}{2}$
- $G \setminus K$ contains an element of G of order 2
- If $i \in \{0, \dots, n-1\}$ satisfies $i^2 \equiv 1 \pmod{n}$ implies $i \equiv \pm 1 \pmod{n}$

Then $G \cong D_{2n}$.

Proof Let G be a group satisfying the above requirements. Then K is a normal subgroup of G . Let x be an element of $G \setminus K$ of order 2. Let $H = \langle x \rangle$. Then $H \cap K = \{1_G\}$. This implies that $|HK| = |H||K| = 2p = |G|$. Thus have that $G = HK$. Let $\phi : H \rightarrow \text{Aut}(K)$ be the homomorphism defined by $\phi_h(k) = hkh^{-1}$ for each $h \in H$. Then from groups and rings theorem 5.3.5 we have that

$$G \cong H \rtimes_{\phi} K$$

The next step is to show that $\phi_h(k) = k^{-1}$ for all $k \in K$. Write $K = \langle y \rangle$ where $|y| = n$. Then $xyx^{-1} = y^i$ for some $0 \leq i < n$. This means that

$$\begin{aligned} y &= x^2 y x^{-2} & (x^2 = 1) \\ &= x y^i x^{-1} \\ &= (x y x^{-1})^i \\ &= y^{i^2} \end{aligned}$$

It follows that $y^{i^2-1} = 1_G$. Since $n = |y|$ we have that $i \equiv \pm 1 \pmod{n}$. Since $0 \leq i < n$, we have that $i = 1$ or $n-1$. If $i = 1$, then $xyx^{-1} = y$ means that G is abelian, which is a contradiction. So we must have $xyx^{-1} = y^{-1}$. Thus the homomorphism ϕ_h is given by $\phi_1(k) = k$ and $\phi_h(k) = k^{-1}$ for all $k \in K$.

By the above proposition, we must have $G \cong H \rtimes_{\phi} K \cong D_{2n}$. ■

Some particular integers satisfying the third condition above include

- If $n = 6$, then by inspection it is easy to see that $i^2 \equiv 1 \pmod{6}$ if and only if $i = 1$ or 5
- If $n = p$ a prime, then $\mathbb{Z}/p\mathbb{Z}$ is a field and has no zero divisor. We can then factorize the expression into $(i-1)(i+1) \equiv 0 \pmod{p}$ so that $i = 1$ or $p-1$.
- If $n = p^2$ a square of prime, the case $p = 2$ is obvious by inspection. Namely $i = 1$ or 3 . If p is odd, p^2 divides $(i-1)(i+1)$. Notice that p cannot divide both $i+1$ and $i-1$ else p divides $i+1+i-1 = 2i$, which is a contradiction. So p divides either $i-1$ or $i+1$. Thus $i = 1$ or p^2-1 .

These are examples of quadratic residues in number theory. Notice that the above examples all have two

solutions. This is similar to the case where a quadratic can have at most two solutions.

6.3 Groups of Order $p, p^2, 2p, 2p^2$

In this section we classify all groups of order $p, p^2, 2p, 2p^2, pq$ for p a prime. We also give some result on the case pq where p and q are distinct.

Proposition 6.3.1 If $|G| = p$ is prime, then $G \cong C_p$.

Proof Let $1 \neq g \in G$. By Lagrange's theorem, $|g| \mid p$ and thus $|g| = p$ and $G = \langle g \rangle$. ■

Proposition 6.3.2 If $|G| = p^2$ where p is prime, then $G \cong C_{p^2}$ or $G \cong C_p \times C_p$.

Proof We know by proposition 6.1.1 that all groups of order p^2 are abelian. The results then follows by the fundamental theorem of finite abelian groups. ■

Proposition 6.3.3 If $|G| = 2p$ with p an odd prime, then either $G \cong C_{2p}$ or $G \cong D_{2p}$.

Proof Let G be a group of order $2p$. If G is abelian, then we must have $G \cong C_{2p}$ by the fundamental theorem of finite abelian groups. Now assume that G is non-abelian. Let $P \in \text{Syl}_p(G)$. Since $|\text{Syl}_p(G)|$ divides $\frac{|G|}{|P|} = 2$ by corollary 5.2.1, together with $|\text{Syl}_p(G)| \equiv 1 \pmod{p}$, we must have that $|\text{Syl}_p(G)| = 1$. This means that P is a normal subgroup of G . Since p is odd, all elements of G of order 2 lies in $G \setminus P$. Now since $\mathbb{Z}/p\mathbb{Z}$ is a field, the only solutions to the equation $x^2 - 1$ in $\mathbb{Z}/p\mathbb{Z}$ are congruent to 1 modulo p . It then follows from theorem 6.2.3 that $G \cong D_{2p}$ as required. ■

Definition 6.3.4 (Generalized Dihedral Group) Let p be an odd prime. Let $H = C_2 = \langle x \rangle$ and let $K = C_p \times C_p$. Let $\phi : H \rightarrow \text{Aut}(K)$ be the inversion homomorphism. Define the generalized dihedral group of order $2p^2$ to be the semidirect product

$$\text{GD}_{2p^2} = H \rtimes_{\phi} K$$

Proposition 6.3.5 Let G be a group of order $2p^2$ where p is odd. Then G is isomorphic to one of the following groups:

- The cyclic group C_{2p^2}
- $C_p \times C_{2p}$
- $C_p \times D_{2p}$
- The dihedral group D_{2p^2}
- The generalized dihedral group GD_{2p^2}

Proof Suppose that G is abelian. Then by the fundamental theorem of finite abelian groups, G is isomorphic to either C_{2p^2} or $C_{2p} \times C_p$. So suppose that G is not abelian. Let $K \in \text{Syl}_p(G)$ and $H = \langle x \rangle \in \text{Syl}_2(G)$ with $|x| = 2$. Then $|K| = p^2$, so $[G : K] = 2$ and that $K \trianglelefteq G$. By proposition 6.2.2, we have that K is isomorphic to either C_{p^2} or $C_p \times C_p$.

Case 1: $K \cong C_{p^2}$.

It is clear that G now satisfies the first two conditions of theorem 6.1.3. It remains to show the last. Suppose that $i \in \{0, \dots, p^2 - 1\}$ is such that $i^2 \equiv 1 \pmod{p^2}$. Then p^2 divides $(i - 1)(i + 1)$. Since p is odd, either p does not divide $i - 1$ or p does not divide $i + 1$. This means that either p^2 divides $i - 1$ or p^2 divides $i + 1$. Since $0 \leq i < p^2$, the only possibilities are either $i = 1$ or $i = p^2 - 1$, and so we conclude that

$$G \cong D_{2p^2}$$

in this case.

Case 2: $K \cong C_p \times C_p$.

In this case, G , satisfies all the conditions of Fitting's lemma so that

$$G \cong (H \rtimes_{\phi} K) \times C_K(x)$$

where ϕ is the inversion homomorphism. By Lagrange's theorem, we must have $|C_K(x)| = 1$ or p or p^2 . However, since $G = \langle K \cup \{x\} \rangle$, we see that $C_K(x)$ cannot have order p^2 . Otherwise, G would be abelian since K is abelian. Note that $|[K, x]| = \frac{|K|}{|C_K(x)|}$ by Fitting's lemma. There are now two cases.

Case 2(a): $|C_K(x)| = 1$.

We then have that $G \cong H \rtimes_{\phi} K$ since $[K, x] = K$ and $C_K(x) = \{1\}$. Combining the fact that $K \cong C_p \times C_p$, we have that

$$G \cong \text{GD}_{2p^2}$$

Case 2(b): $|C_K(x)| = p$.

In this case, we have $|C_K(x)| = p = [K, x]$. Thus $H \rtimes_{\phi} [K, x]$ is a non-abelian group of order $2p$. Hence $H \rtimes_{\phi} [K, x] \cong D_{2p}$ by proposition 6.2.3. Since $C_K(x) \cong C_p$ by proposition 6.2.1, we deduce that

$$G \cong D_{2p} \times C_p$$

which completes the proof. ■

Proposition 6.3.6 Let p and q be distinct primes with $p < q$ and p does not divide $q - 1$. Let G be a group of order pq . Then $G \cong C_{pq}$.

Proof By corollary 5.2.1, we know that $n_p(G) | q$. By Sylow's theorem, we have that $n_p(G) \equiv 1 \pmod{p}$. If $n_p(G) = q$, then p divides $q - 1$ which is a contradiction. So $n_p(G) = 1$. By a similar argument and the fact that $q > p$, we have that $n_q(G) = 1$. Therefore we have that G has a normal p -subgroup, say H and a normal q -subgroup, say K . By Lagrange's theorem, we must have that $H \cap K = \{1\}$. This means that $|HK| = |H||K|$. By proposition 5.2.5 in Groups and Rings, we have that $G = HK$. Hence we have that $G \cong H \times K$ by proposition 5.2.5 in Groups and Rings. Let x and y be generators of H and K respectively. Then x and y commute since $H \cap K = \{1\}$. So $|xy| = |x||y| = pq$ by proposition 1.1.9 in Groups and Rings. Thus $G = \langle x, y \rangle$. ■

6.4 Groups of Order up to 12

The Klein Four Group is the only other group of order 4 that is not isomorphic to the cyclic group.

Definition 6.4.1 (The Klein Four Group K_4) The Klein four group of order 4 is defined to be the group

$$K_4 = \{1, a, b, c\}$$

where multiplication is defined by $a^2 = b^2 = c^2 = 1$, $ab = ba = c$, $ac = ca = b$ and $bc = cb = a$.

Treating c in the definition as ab makes it easy to see that K_4 is in fact D_4 in disguise.

Lemma 6.4.2 The Klein four group K_4 is isomorphic to D_4 but not isomorphic to C_4 .

Definition 6.4.3 (Quaternion Group) Let i, j, k be indeterminates and consider the set

$$Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$$

together with a binary operation $\cdot : Q_8 \times Q_8 \rightarrow Q_8$ defined by

- $1 \cdot g = g \cdot 1 = 1$ and $(-1) \cdot g = g \cdot (-1) = -g$ for all $g \in Q_8$
- $i \cdot j = k, j \cdot k = i$ and $k \cdot i = j$

- $j \cdot i = -k, k \cdot j = -i$ and $i \cdot k = -j$
- $(\pm 1)^2 = 1, (\pm i)^2 = (\pm j)^2 = (\pm k)^2 = -1$

Then $(Q_8, \cdot)$ is called the Quaternion group.

Proposition 6.4.4 The following are properties of the quaternion group Q_8 .

- $Z(Q_8) = \{\pm 1\}$
- Q_8 has precisely 1 element of order 2, which is -1
- Q_8 has precisely 6 elements of order 4, which is $\pm i, \pm j$ and $\pm k$
- $Q_8 = \langle i, j \rangle = \langle j, k \rangle = \langle k, i \rangle$

Theorem 6.4.5 Any group of order 8 is isomorphic to one of the following groups:

- The Dihedral group D_8
- The Quaternion group Q_8
- The Cyclic group C_8
- $C_4 \times C_2$
- $C_2 \times C_2 \times C_2$

Proof Let G be a group of order 8. Suppose that G is abelian. Then G is isomorphic to one of $C_2 \times C_2 \times C_2$ or $C_4 \times C_2$ or C_8 by the fundamental theorem of finitely generated abelian groups. So suppose that G is nonabelian. Then G has no element of order 8 else $G \cong C_8$. Moreover G must have at least one nontrivial element of order not equal to 2 else G is abelian. So say $u \in G$ has order larger than 2. By Lagrange's theorem we must have $|u| = 4$. Let $H = \langle u \rangle$. Let $v \in G \setminus H$ with $|v|$ minimal. Then either $|v| = 2$ or 4. There are two cases.

Case 1: $|v| = 2$

Suppose that $|v| = 2$. Then G satisfies the three hypotheses for the Dihedral group by proposition 4.1.2 and thus $G \cong D_8$.

Case 2: $|v| = 4$

Suppose that $|v| = 4$. Then since $v \notin H$ and $[G : H] = 2$, we must have $G/H = \{H, vH\}$. Since $v \in H$ has minimal order, this means that all elements in vH has order 4. Since furthermore $u, u^{-1} \in H$ both has order 4, there are precisely 6 elements of order 4 and one element of order 2 (u^2 has order 2). It follows that for x an element of order 6, $x^2 = u^2$ since u^2 has order 2. Some algebra shows that $u^2x = xu^2$ which means that $u^2 \in Z(G)$. We also have $u^2x = x^3 = x^{-1}$ for all x with order 6.

Side claim: $vuv^{-1} = u^{-1}$

Clearly $G = \langle u, v \rangle$ since $v \notin H = \langle u \rangle$, and $|H| = 4$. Moreover, $H \trianglelefteq G$ since $[G : H] = 2$. Thus $vuv^{-1} \in H$. If $vuv^{-1} = u$, the v and u commute, thus G is abelian which is a contradiction. u^3 is then the only element of order 4 in H and we know that $|vuv^{-1}| = |u|$. This means that $vuv^{-1} = u^{-1}$.

Continuing the proof, let $w = uv$. We have that

$$\begin{aligned} w &= uv \\ &= vuvv^{-1} \\ &= vu^{-1} \in vH \end{aligned} \quad (vuv^{-1} = u^{-1})$$

We already know that u, u^{-1}, v, v^{-1} are distinct elements since they lie in different cosets. But also now w, w^{-1} is not any of the four. Indeed $w \in vH$ thus w, w^{-1} will not be equal to u, u^{-1} . But if $w = v$, then this implies $u = 1$, a contradiction. Similarly, if $w = v^{-1}$, then $v^2 = u$ which is also a contradiction since $|v^2| = 2$ while $|u| = 4$.

Thus we must have that

$$G = \{1, u^2, u^{\pm 1}, v^{\pm 1}, w^{\pm 1}\} = \{1, u, u^2, u^3, v, u^2v, w, u^2w\}$$

Notice that

$$\begin{aligned}
 uv &= vv^{-1}uv \\
 &= vu^{-1} \\
 &= vu^3 \\
 &= (vu)(u^2) \\
 &= (u^2)(vu)
 \end{aligned}$$

A similar argument show that $gh = u^2hg$ for all $g, h \in \{u, v, w\}$. Using the fact that $v^2 = w^2 = u^2 \in Z(G)$, we see that

- $u^2g = gu^2$ for all $g \in G$
- $uv = w, vw = vuv = u^2v^2u = u$ and $wu = uvu = u^2vu^2 = v$
- $vu = u^2w, wv = u^3, uw = u^2v$
- $(\pm 1)^2 = 1$ and $(\pm g)^2 = u^2$ for all $g \in \{u, u^3, v, u^2v, w, u^2w\}$

It is easy to see that the Cayley diagram can be formed from these relations which shows that this group is isomorphic to Q_8 . ■

Definition 6.4.6 (Dicyclic Group of Order 12) Let $C_4 = \langle h \rangle$. Define $\phi : C_4 \rightarrow \text{Aut}(C_3)$ to be the inversion homomorphism for. Define the dicyclic group of order 12 to be

$$\text{Dic}_{12} = C_4 \rtimes_{\phi} C_3$$

Theorem 6.4.7 Let G be a group of order 12. Then G is isomorphic to one of the following groups:

- The cyclic group C_{12}
- $C_6 \times C_2$
- Dic_{12}
- The dihedral group D_{12}
- The alternating group A_4

Proof When G is abelian, the fundamental theorem of finite abelian group tells us that G is isomorphic to either C_{12} or $C_6 \times C_2$. So suppose that G is non-abelian. We split into two cases.

Case 1: G contains an element of order 6.

Let $|a| = 6$ and $K = \langle a \rangle$.

Case 1(a): $G \setminus K$ has an element of order 2.

Then G satisfies the first two hypothesis of theorem 6.1.3. It remains to show that the final condition is satisfied. But it is clear that $2^2, 3^2, 4^2$ is congruent to 4, 3, 4 respectively modulo 6. So i^2 modulo 6 is 1 if and only if $i = 1$ or 5. So we can conclude that in this case, $G \cong D_{12}$.

Case 1(b): No element of $G \setminus K$ is of order 2.

Let $H \in \text{Syl}_2(G)$. Then H is not a subgroup of K since K has two elements of order 2 and $|H| = 4$. If every non-identity element of H has order 2, then $H \setminus K$ would consist of elements of order 2, contrary of our assumption that $G \setminus K$ has no elements of order 2. Thus H must have an element of order 4 and thus $H \cong C_4$. Now let $K_1 = \langle a^2 \rangle$. Then $|K_1| = 3$. Thus $K_1 \in \text{Syl}_3(G)$.

Claim: K_1 is normal. We want to show that $ga^2g^{-1} \in K_1$ and $ga^4g^{-1} \in K_1$. Since K is normal, we have that $ga^2g^{-1} = a^i$ for $i \in \{0, \dots, 5\}$. But $(ga^2g^{-1})^3 = 1$ implies that $a^{3i} = 1$. This means that $3i$ is a multiple of 6. This means that i is a multiple of 2. Thus $ga^2g^{-1} \in K_1$. A similar method shows that $ga^4g^{-1} \in K_1$. Thus K_1 is normal.

We now have that $H \leq G$, $K_1 \trianglelefteq G$, $H \cap K_1 = \{1\}$ by Lagrange's theorem and $|HK_1| = |H| = |K_1| = |G|$ implies that $HK_1 = G$ by proposition 5.2.5 in Groups and Rings. Using proposition 5.3.5 in Groups and Rings, we have that $G \cong H \rtimes_{\phi} K$ where $\phi : H \text{Aut}(K)$ is given by $\phi_h(k) = hkh^{-1}$. Suppose that $H \cong C_4 = \langle h \rangle$ and $k \in K \setminus \{1_G\}$. Then $G = \langle h, k \rangle$ so $hkh^{-1} \neq k$ since G is non-abelian. Since $hkh^{-1} \in K_1$ and $|hkh^{-1}| = |k|$, we must have that $hkh^{-1} = k^{-1}$. This means ϕ is the inversion homomorphism and that G is just the dicyclic group of order 12 by definition.

Case 2: G has no element of order 6.

By Cauchy's theorem, we may choose an element of order 3. Since $x \in C_G(x)$ and G has no element of order 6, we see that 2 does not divide $|C_G(x)|$. Indeed if it did, then we could choose an element of order 2 in $C_G(x)$ by Cauchy's theorem. But then $|xy| = 6$ by proposition 1.3.6 in Groups and Rings. Thus we must have that $|C_G(x)| = 3$ by Lagrange's theorem. By the orbit stabilizer theorem, we have that $|Cl(x)| = [G : C_G(x)] = 4$. Since G has precisely $n_3(G) \times (3 - 1)$ elements of order 3 by corollary 5.3.1, we deduce that $n_3(G) > 1$. In particular, H is not a normal subgroup of G .

Finally, let $H \in \text{Syl}_3(G)$ and consider the action of G on $\frac{G}{H}$ by left multiplication. Let K be the kernel of this action. Then $K \leq H$ by ???. But H is not normal in G . So we must have $|K| = 1$ since $|H| = 3$. Thus $G \cong \frac{G}{K}$ is isomorphic to a subgroup of S_4 . But the only subgroup of S_4 of order 12 is A_4 . Thus $G \cong A_4$. ■

We provide a table for the classification of finite groups up to order 12:

Group Order	Groups
1	*
2	C_2
3	C_3
4	$C_4, C_2 \times C_2$
5	C_5
6	C_6, S_3
7	C_7
8	$C_8, C_4 \times C_2, C_2 \times C_2 \times C_2, D_8, Q_8$
9	$C_9, C_3 \times C_3$
10	C_{10}, D_{10}
11	C_{11}
12	$C_{12}, C_6 \times C_2, A_4, D_{12}, \text{Dic}_{12}$

6.5 Unique Simple Group of Order 60

As an application of the classification of small groups, together with the Sylow theorems, we prove that the unique group of order 60 is A_5 .

Theorem 6.5.1 Let G be a simple group of order 60. Then $G \cong A_5$.

Proof

Step 1: G does not contain a proper subgroup of index $n \leq 4$. Assume the contrary. Let $H \leq G$ with $1 \neq [G : H] \leq 4$. Then G acts non-trivially by left multiplication on the set of left cosets G/H of H in G , and $|G/H| = n$ for some $2 \leq n \leq 4$. Since G is simple, the kernel of this action is trivial. Thus G is isomorphic to a subgroup of S_n . Since $2 \leq n \leq 4$, we have that $|S_n| \leq 4! = 24$. But $|G| = 60$ so that this contradicts Lagrange's theorem.

Step 2: G contains a subgroup of index 5.

Assume the contrary. Let $P \in \text{Syl}_2(G)$. By Sylow's theorem, we have that $n_2(G)$ is odd and $n_2(G)$ divides 15. Moreover $n_2(G) \neq 1$ since G is simple. We also know that $n_2(G) = \frac{|G|}{|N_G(P)|}$ so by step

1, $n_2(G) = 5$ or 15. If $n_2(G) = 5$ then $\frac{|G|}{|N_G(P)|} = 5$, a contradiction.

So suppose $n_2(G) = 15$. Let $g \in G \setminus N_G(P)$. Let $H = \langle P \cup gPg^{-1} \rangle$. Then H contains at least two Sylow 2-subgroups namely P and gPg^{-1} . So $n_2(H)$ being an odd number, is at least 3. Since 4 divides $|H|$, we deduce that $|H|$ is divisible by $4n_2(H)$. By step 1, we have either $H = G$ or $|H| < \frac{|G|}{4} = 15$. Thus we must have either $H \neq G$ and $n_2(H) = 3$ or $H = G$. If $H \neq G$ and $n_2(H) = 3$, then 12 divides $|H|$ so we must have $|H| = 12$ by step 1. Thus $[G : H] = 5$ as needed.

Now assume that for all $g \in G \setminus N_G(P)$, we then have $G = \langle P \cup gPg^{-1} \rangle$. Assume that $x \in P \cap gPg^{-1}$, then x is centralized by both P and gPg^{-1} , since groups of order 4 are abelian. Thus x is centralized in G since $G = \langle P \cup gPg^{-1} \rangle$. Then $x \in Z(G)$. Since G is simple, we have $Z(G) = \{1\}$ and so $x = 1$ and we conclude that $P \cap gPg^{-1} = \{1\}$ for all $g \in G$. Now

$$F_2(G) = \bigcup_{P \in \text{Syl}_2(G)} P \setminus \{1_G\}$$

by consequences of the Sylow theorems. By the above work, this is a disjoint union. Thus $|F_2(G)| = n_2(G) \times 3 = 45$. By Sylow's theorem, we have that $n_5(G) \equiv 1 \pmod{5}$ and $n_5(G)$ divides 12. Since G is simple, we have $n_5(G) > 1$ so that $n_5(G) = 6$. But then $|F_5(G)| = 6 \times 4 = 24$. This means that $|G| > |F_2(G)| + |F_5(G)| = 69$, a contradiction. This completes the proof of step 2.

Step 3: Final result.

Let $H \leq G$ be a subgroup of index 5. Then G acts non-trivially by left multiplication on the set of left cosets G/H of H in G . Since G is simple, the kernel of this action is trivial, and thus G is isomorphic to a subgroup of S_5 . But the only subgroup of S_5 of order 60 is A_5 , which completes the proof. ■

7 Series of Subgroups

7.1 Series of Subgroups

Definition 7.1.1 (Subnormal Series) Let G be a group. A subnormal series of a group G is a sequence of subgroups, each a normal subgroup of the next one. This is written as

$$\{1\} = G_0 \triangleleft G_1 \triangleleft \cdots \triangleleft G_n = G$$

In this case we say that the series has length n .

Definition 7.1.2 (Types of Subnormal Series) Let G be a group and

$$\{1\} = G_0 \triangleleft G_1 \triangleleft \cdots \triangleleft G_n = G$$

a subnormal series of G . Then the subnormal series can have additional properties.

- The series is normal if each $G_i \triangleleft G$ for $0 \leq i \leq r$
- The series is a composition series if G_{i+1}/G_i is simple for $0 \leq i \leq r-1$
- The series is a central series if it is a normal series in which $G_{i+1}/G_i \leq Z(G/G_i)$ for $0 \leq i \leq r-1$

7.2 Composition Series

Lemma 7.2.1 Let G be a finite group. Let N be a normal subgroup of G . Suppose that

$$\{1_G\} = N_0 \triangleleft N_1 \triangleleft \cdots \triangleleft N_r = N$$

and

$$\{1_{G/N}\} = \frac{X_0}{N} \triangleleft \frac{X_1}{N} \triangleleft \cdots \triangleleft \frac{X_s}{N} = \frac{G}{N}$$

are composition series for N and $\frac{G}{N}$ respectively. Set $G_i = N_i$ for $0 \leq i \leq r$ and $G_i = X_i$ for $r+1 \leq i \leq r+s$. Then

$$G_0 \triangleleft \cdots \triangleleft G_s$$

is a composition series of G .

Proof It is clear that we can write the composition series of $\frac{G}{N}$ as quotient groups by the correspondence theorem: Suppose we write $\overline{G}_i$ for the composition series of $\frac{G}{N}$. Fix $0 \leq i \leq s$. By the correspondence theorem, there exists a subgroup X_i of G containing N such that $\overline{G}_i = \frac{X_i}{N}$.

Since $\overline{G}_i$ is a normal subgroup of $\overline{G}_{i+1}$, it follows that X_i is a normal subgroup of X_{i+1} . By the third isomorphism theorem, we have

$$\frac{\overline{G}_{i+1}}{\overline{G}_i} = \frac{X_{i+1}/N}{X_i/N} \cong \frac{X_{i+1}}{X_i}$$

so that $\frac{X_{i+1}}{X_i}$ is simple. Now notice that $X_s = G$ and $N_r = N = X_0$. Set $G_i = N_i$ for $0 \leq i \leq r$ and set $G_i = X_{i-r}$ for $r+1 \leq i \leq r+s$. Then $G_0 \triangleleft \cdots \triangleleft G_{r+s}$ is a composition series for G . ■

Proposition 7.2.2 Every finite group has a composition series.

Proof We induct on $|G|$. If $|G| = 1$, there is a trivial composition series. So assume that $|G| > 1$, and that every group of order less than $|G|$ has a composition series. If G is simple, we are done. So assume G is not simple. Then G has a normal subgroup N with $\{1_G\} \neq N \neq G$. Then $|N| < |G|$ and $\frac{G}{N} < |G|$, so our inductive hypothesis implies that N has a composition series $N_0 \triangleleft \cdots \triangleleft N_r$ and $\frac{G}{N}$ has a composition series $\overline{G}_0 \triangleleft \cdots \triangleleft \overline{G}_{r+s}$. The results follow by the above lemma. ■

Definition 7.2.3 (Equivalent Composition Series) Let G be a group. Two composition series of G , namely $A_0 \triangleleft \cdots \triangleleft A_r$ and $B_0 \triangleleft \cdots \triangleleft B_s$ is said to be equivalent if $r = s$ and there exists a bijection

$$f : \{A_i/A_{i-1} | 1 \leq i \leq r\} \rightarrow \{B_j/B_{j-1} | 1 \leq j \leq s\}$$

such that $A_i/A_{i-1} \cong f(A_i/A_{i-1})$ for $1 \leq i \leq r$.

Theorem 7.2.4 (The Jordan–Hölder Theorem) Let $A_0 \triangleleft \cdots \triangleleft A_r$ and $B_0 \triangleleft \cdots \triangleleft B_s$ be two compositions series of a finite group G . Then they are equivalent.

Proof Without loss of generality, let $r \leq s$. We induct on r . If $r = 0$, then $G = \{1_G\}$ and the result is clear. Assume that $r > 0$. There are two cases.

Case 1: $A_{r-1} = B_{s-1}$.

Then $A_0 \triangleleft \cdots \triangleleft A_{r-1}$ and $B_0 \triangleleft \cdots \triangleleft B_{s-1}$ are two composition series of $A_{r-1} = B_{s-1}$ of lengths $r - 1$ and $s - 1$. By induction hypothesis, they are equivalent and hence so are the two composition series of G .

Case 2: $A_{r-1} \neq B_{s-1}$.

Consider $A_{r-1}B_{s-1}$. Since A_r and B_s are proper normal subgroups of G , we have $B_{s-1} < A_{r-1}B_{s-1} \trianglelefteq G$. It follows from the third isomorphism theorem that $A_{r-1}B_{s-1}/B_{s-1}$ is a normal subgroup of G/B_{s-1} . Since $B_{s-1} < A_{r-1}B_{s-1}$, it is in fact a non-trivial normal subgroup by the correspondence theorem. Since G/B_{s-1} is simple, we have $A_{r-1}B_{s-1}/B_{s-1} = G/B_{s-1}$ so that $A_{r-1}B_{s-1} = G$ by the correspondence theorem. Let $D = A_{r-1} \cap B_{s-1}$. By the second isomorphism theorem, we have

$$\frac{G}{A_{r-1}} = \frac{A_{r-1}B_{s-1}}{A_{r-1}} \cong \frac{B_{s-1}}{A_{r-1} \cap B_{s-1}} = \frac{B_{s-1}}{D}$$

and as G/A_{r-1} is simple, so is B_{s-1}/D . Similarly, by applying the second isomorphism theorem again, we have

$$\frac{G}{B_{s-1}} = \frac{A_{r-1}B_{s-1}}{B_{s-1}} \cong \frac{A_{r-1}}{A_{r-1} \cap B_{s-1}} = \frac{A_{r-1}}{D}$$

and as G/B_{s-1} is simple, so is A_{r-1}/D . Now let

$$\{1_G\} = D_0 \triangleleft \cdots \triangleleft D_t = D$$

be a composition series for D . Then

$$\{1_G\} = D_0 \triangleleft \cdots \triangleleft D_t = D \triangleleft A_{r-1} \triangleleft G$$

and

$$\{1_G\} = D_0 \triangleleft \cdots \triangleleft D_t = D \triangleleft B_{s-1} \triangleleft G$$

are both composition series of G .

Since $r \leq s$, we can apply the first case to the get that

$$A_0 \triangleleft \cdots \triangleleft A_r \quad \text{and} \quad D_0 \triangleleft \cdots \triangleleft D_t = D \triangleleft A_{r-1} \triangleleft G$$

are equivalent. In particular, we have that $r = t + 2$ and hence $D_0 \triangleleft \cdots \triangleleft D_t = D \triangleleft B_{s-1} \triangleleft G$ has length $t + 2 = r$. Thus we can apply case 1 again to conclude that

$$B_0 \triangleleft \cdots \triangleleft B_s \quad \text{and} \quad D_0 \triangleleft \cdots \triangleleft D_t = D \triangleleft B_{s-1} \triangleleft G$$

are equivalent. Finally, we have seen that $G/A_{r-1} \cong B_{s-1}/D$ and $A_{r-1}/D \cong G/B_{s-1}$. Thus

$$D_0 \triangleleft \cdots \triangleleft D_t = D \triangleleft A_{r-1} \triangleleft G \quad \text{and} \quad D_0 \triangleleft \cdots \triangleleft D_t = D \triangleleft B_{s-1} \triangleleft G$$

are equivalent so that we conclude. ■

We have thus proved that the following definition of is an invariant of a group.

Definition 7.2.5 (Composition Factors) Let G be a group and

$$\{1_G\} = G_0 \triangleleft \cdots \triangleleft G_r = G$$

be a composition series of G . We say that the simple groups $\frac{G_{i+1}}{G_i}$ for $0 \leq i < r$ are called the composition factors of G .

7.3 Soluble Groups and Derived Series

Soluble group, while interesting in its own right, has two main applications: To the study of Galois theory and to the study of composition series. We will focus on its relations with composition series and leave the relation to Galois theory in Fields and Galois Theory.

Definition 7.3.1 (Soluble Groups) A group G is said to be soluble if it is either trivial, or its composition factors are all cyclic groups of prime order.

Notice that all abelian groups are soluble. Indeed every quotient group of G is abelian and so every composition factors of G are abelian simple groups, which must be cyclic groups of prime order.

Lemma 7.3.2 Let G be a finite group. Let N be a normal subgroup of G . Then G is soluble if and only if both N and $\frac{G}{N}$ are soluble.

Proof Write $\text{CF}(G)$ the composition factors of G . By the Jordan–Hölder theorem and lemma 7.2.1, we have

$$\text{CF}(G) = \text{CF}(N) \cup \text{CF}(G/N)$$

Thus G is soluble if and only if both N and G/N are soluble. ■

Definition 7.3.3 (Derived Series) Let G be a group. Define $G^{(1)} = [G, G]$ and inductively, $G^{(n+1)} = [G^{(n)}, G^{(n)}]$ for integers $n > 0$. We call the descending series

$$G = G^{(0)} \geq G^{(1)} \geq \cdots \geq G^{(n)} \geq \cdots$$

the derived series of G .

Lemma 7.3.4 Let G be a group and $G = G^{(0)} \geq G^{(1)} \geq \cdots \geq G^{(n)} \geq \cdots$ its derived series. Then the following are true:

- $(G^{(n)})^{(m)} = G^{(n+m)}$
- $H^{(n)} \leq G^{(n)}$ if $H \leq G$

Proof

- We proceed by induction on m . When $m = 0$ the results are clear. If $(G^{(n)})^{(m-1)} = G^{(n+m-1)}$, then we have

$$\begin{aligned} (G^{(n)})^{(m)} &= [(G^{(n)})^{m-1}, (G^{(n)})^{m-1}] \\ &= [G^{(n+m-1)}, G^{(n+m-1)}] \\ &= G^{(n+m)} \end{aligned}$$

and so we are done.

- Similarly by induction, we have

$$\begin{aligned} H^{(n)} &= [H^{(n-1)}, H^{(n-1)}] \\ &\leq [G^{(n-1)}, G^{(n-1)}] \\ &= G^{(n)} \end{aligned}$$

And so we conclude. ■

We focus the rest of the chapter to the case where G is finite.

Theorem 7.3.5 Let G be a finite group. The following are equivalent characterizations for solubility.

- G is soluble.
- $G^{(n)} = 1$ for some $n \geq 0$ in the derived series of G
- G has a normal series with abelian factors.

Proof

- (1) $\implies$ (2): Suppose that G is soluble. We proceed by induction on the order of G . When $|G| = 1$ the results are clear. Suppose that $|G| > 1$. Let $N = [G, G]$. Since N is normal and G is soluble, N is soluble by lemma 7.2.1. By definition of solubility, G has all composition factors cyclic groups of prime order. In particular, G_{r-1} is a normal subgroup of G with

$$G_r/G_{r-1} = G/G_{r-1}$$

being cyclic of prime order. By proposition 3.4.3, we have that $N \leq G_{r-1}$. In particular, $|N| < |G|$. So $N^{(m)} = 1$ by induction hypothesis. Since $G^{(n)} = [G^{(n-1)}, G^{(n-1)}]$ by definition, it follows that $G^{(m+1)} = 1$ and so we are done.

- (2) $\implies$ (1): Suppose that $G^{(n)} = 1$ for some $n \in \mathbb{N}$. We will again induct on the order of G . This is clearly true if $|G| = 1$. So suppose that $|G| > 1$. Let $N = [G, G]$. If $N = G$ then we have that

$$G^{(n)} = [G, G]^{(n-1)} = G^{(n-1)} = \dots = G$$

which is a contradiction. So we must have $N < G$. Since $N^{(n-1)} = G^{(n)} = 1$, we have that N is soluble by induction hypothesis. Also $G/N = G/[G, G]$ is abelian so that both N and G/N is soluble. Hence G is soluble by lemma 7.4.2. Thus we are done. ■

Corollary 7.3.6 Let G be a finite soluble group and $H \leq G$. Then H is soluble.

Proof Suppose that G is soluble. Then $G^{(n)} = 1$ for some $n \in \mathbb{N}$. Then since $H^{(n)} \leq G^{(n)}$, we have $H^{(n)} = 1$ and so H is soluble. ■

Theorem 7.3.7 Let p be prime and $n \in \mathbb{N}$. Let G be a finite group of order p^n . Then G is soluble, and all composition factors of G are isomorphic to C_p .

Proof We prove the theorem by induction on $|G|$. If $|G| = p$ then G is cyclic of prime order, so G is soluble of composition length 1, and its composition factors are C_p .

Assume that $|G| > p$. The normal subgroup $Z(G)$ is non-trivial in this case. Since $Z(G)$ is abelian, $Z(G)$ is soluble. By induction hypothesis, $G/Z(G)$ is soluble. Thus G is soluble by lemma 7.4.2. By the same lemma, we have that the composition factors of G is the union of the composition factors of $Z(G)$ and $G/Z(G)$. By induction hypothesis all their composition factors are isomorphic to C_p and so we conclude. ■

Theorem 7.3.8 Let G_1, G_2 be soluble groups. Then $G_1 \times G_2$ is soluble.

Proof Recall that $G_1 \times G_2 \cong G_1 \rtimes_{\phi} G_2$ where $\phi : G_2 \rightarrow \text{Aut}(G_1)$ is the trivial homomorphism. Also since $G_1 \trianglelefteq G_1 \times G_2$, and $G_1 \times G_2 / G_1 \cong G_2$, together with G_1, G_2 being soluble implies that $G_1 \times G_2$ is soluble. ■

As usual, this theorem can be applied inductively to see that $G_1 \times \cdots \times G_n$ is soluble given that $G_1, \dots, G_n$ are soluble.

7.4 Nilpotent Groups

Definition 7.4.1 (Nilpotent Groups) Let G be a group. We say that G is nilpotent if G has a central series of finite length.

Definition 7.4.2 (Lower Central Series) Let G be a group. Define the lower central series of G to be the descending series of subgroups

$$G = G_1 \supseteq G_2 \supseteq \cdots \supseteq G_n \supseteq \cdots$$

where $G_{n+1} = [G_n, G]$. We often write $\gamma_n(G) = G_n = [G_{n-1}, G]$ for the n th term in the lower central series of G .

Proposition 7.4.3 Let G be a group. Then G is nilpotent if and only if the lower central series terminates at the trivial subgroup after finitely many steps.

Definition 7.4.4 (Nilpotency Class) Let G be a nilpotent group. Define the nilpotency class of G to be the smallest $n \in \mathbb{N}$ such that G has a central series of length n . Equivalently, the nilpotency class is the maximal of $n \in \mathbb{N}$ such that $\gamma_n(G) \neq 1$.

Proposition 7.4.5 Let G be a nilpotent group. Let $H \leq G$ and $N \trianglelefteq G$. Then the following are true.

- H is nilpotent
- $\gamma\left(\frac{G}{N}\right) = \frac{\gamma_n(G)N}{N}$ for all $n \in \mathbb{N}$. In particular, G/N is nilpotent.

Proposition 7.4.6 Let G be a finite nilpotent group. Then G is soluble.

Proposition 7.4.7 Let G be a finite group of prime power order. Then G is nilpotent.